An Ontological Completion of Geometric Quantum Mechanics

A Preprint

Zayn Blore zayn.blore@ou.ac.uk
www.constraintsurfacedynamics.com

May 10, 2026

Abstract

Decades of work in geometric quantum mechanics and decoherence theory have clarified how quantum behaviour can be represented through symplectic geometry and dynamical structure, yet neither framework by itself explains why measurements yield definite outcomes with Born-form frequencies. This paper proposes a deterministic geometric framework that retains the strengths of those approaches while avoiding stochastic collapse and multiple-world postulates. The result is a unified account in which observed quantum statistics arise from conserved geometric structure rather than from primitive probabilistic axioms.

In Constraint-Surface Dynamics, physical reality is represented by a single trajectory $\omega(t)$ evolving under Hamiltonian flow on a compact symplectic manifold Σ. A measurement context induces a measurable family of outcome basins $\{\Omega_i\}$ on Σ, and within the typicality framework their invariant volume ratios $\mu(\Omega_i)/\mu(\Omega_0)$ determine long-run outcome frequencies. In the quantum-effective sector, combining this deterministic basin structure with the $\mathrm{SU}(n)$-fixed operational measure and the admissible probability functional yields Born-form weights. Outcome definiteness is attributed to single-trajectory evolution together with context-defined basin selection.

The paper presents CSD as an ontological completion of geometric quantum mechanics in a deterministic single-trajectory setting, and states how the framework is intended to remain compatible with Bell, Kochen–Specker, Fine, and PBR constraints. Detailed technical analyses of those constraints are developed in companion CSD notes. Within its present finite-dimensional scope, the framework shows how a measure-preserving geometry can reproduce the statistical structure of quantum theory while preserving a clear criterion of empirical vulnerability. Constraint-Surface Dynamics is therefore proposed as a deterministic geometric foundation for quantum mechanics.

Note: The measure-fixing and qubit-level checks used in the present paper rely on the formal results established in Paper B (2025b). The role of the present manuscript is to supply the ontological, measurement-theoretic, and interpretive structure that integrates those results into a unified geometric framework.

The role of the present manuscript is conceptual and interpretive rather than fully technical. The measure bridge and Born-weight chain are established in Paper B and TN1; mixed states, POVMs, and subsystem reduction are developed in TN4; finite-dimensional post-measurement update and sequential measurement are developed in TN6; and Bell-, Fine-, and PBR-facing structural analyses are developed in TN0, c1, and c2. Accordingly, the purpose of Paper D is to state the ontological picture, the measurement-theoretic interpretation, and the explanatory role of these results within a single deterministic framework. Deeper derivations of the ontic manifold Σ, the projection map π, and the composite projective target are not claimed here as closed first-principles results, but remain part of the wider CSD research programme.

This paper retains the earlier notation $\omega \in \Sigma$ for the ontic microstate and μ_Σ for the ontic invariant measure. In the programme-wide notation standard adopted in TN0, these are written as $x \in \Sigma$ and μ_L, respectively. No substantive difference is intended.

Contents

1 Why Quantum Mechanics Needs Interpretation

Quantum mechanics predicts experimental outcomes with extraordinary precision, yet what the formalism says about physical reality remains unsettled. The standard package combines linear unitary evolution with a rule for extracting measured outcomes. Between preparation and registration, a pure state $|\Psi\rangle$ evolves by Schrödinger dynamics; at registration, the same $|\Psi\rangle$ is used to compute discrete outcome frequencies via $|\langle\phi_i|\Psi\rangle|^2$. The predictive success is unquestioned, but the explanatory gap is clear: how do continuous, deterministic trajectories in state space yield single, definite outcomes recorded by macroscopic devices (Bell 1990; Maudlin 1995; Schlosshauer 2007)?

Unitary dynamics on Hilbert space, or equivalently Hamiltonian flow on the projective manifold $\mathbb{CP}^{n-1}$, is deterministic and preserves invariant measure structure. Yet observers register one eigenvalue per run. Decoherence theory explains why interference terms become practically unobservable in large environments and why certain pointer structures are stable, but it does not identify which specific outcome occurs in a single trial, nor why the Born weights take the squared-amplitude form without further assumption (Zurek 2003; Schlosshauer 2007). This is the measurement problem in its minimal form: reconcile deterministic flow with outcome selection and weighting.

Calling the projection postulate a pragmatic rule leaves open what the wave function represents and when the rule applies. If $|\Psi\rangle$ is a complete description of reality, then multiplicity of outcomes seems unavoidable; if $|\Psi\rangle$ is only a tool for inference, one must supply what is described. An interpretation is needed to state, in physical terms, what the mathematical objects denote, which dynamical principles are fundamental, and why laboratory observations are single-outcome events. Without this, the formalism is operationally effective but conceptually incomplete.

It is tempting to treat the gap as merely semantic. That view underestimates its reach:

1. **Technology and quantum engineering:** Quantum control, error correction, and metrology depend on when coherence is physically available and how it is lost. A clear account of isolation, partitioning, and outcome registration informs device boundaries, noise models, and fault-tolerant architectures (Nielsen and Chuang 2010). Clarity about what constitutes a measurement shapes experimental design and data interpretation.
2. **Information theory:** Resource theories for coherence, entanglement, and asymmetry presuppose a division between accessible operations and outcome readout. Interpreting the readout as revelation of a pre-existing property versus stochastic generation leads to distinct conceptual, and sometimes mathematical, treatments (Horodecki and Oppenheim 2013).

3. **Cosmology and quantum gravity:** In closed systems without external observers, one cannot appeal to an external classical domain. A consistent account of outcomes and probabilities must be formulated internally. This touches the status of initial conditions, the arrow of time, and the emergence of classical spacetime structures (Gell-Mann and Hartle 1993).
4. **Foundational consistency checks:** No-go theorems constrain hidden variables, noncontextual value assignments, and superluminal influences. An interpretation should make explicit how it adheres to no-signalling, accommodates contextuality where required, and fits within relativistic causal structure (Bell 1964; Kochen and Specker 1967; Fine 1982).

These are not optional clarifications. They determine what counts as a legitimate model of devices, experiments, and large-scale systems.

Any serious proposal must meet four minimal criteria:

- **Definite outcomes:** Each experimental run ends with one record in the laboratory frame. The mechanism for selecting that record must be specified without hand-waving.
- **Origin of the Born weights:** The rule $|\langle\phi_i|\Psi\rangle|^2$ should follow from structural features of the theory rather than be appended as a separate axiom.
- **Observer neutrality:** Human observers and macroscopic apparatus are physical systems. The account must embed registration within dynamics, not rely on an observer-system split defined by fiat.
- **Empirical equivalence with standard quantum mechanics:** Predictions must match orthodox calculations where those are confirmed.

Casting pure states as points on Σ equipped with symplectic structure and the Fubini–Study metric isolates what is kinematically fixed by symmetry and what is added by interpretation. Hamiltonian flow on $\mathbb{CP}^{n-1}$ encodes unitary evolution; $\mathrm{SU}(n)$ invariance singles out a unique operational volume form on Σ. These kinematic facts are rigorous and interpretation-independent. The open question is what the points, flows, and invariant volumes mean physically. If they are only calculational devices, the measurement problem persists. If they represent elements of reality, one must explain how measurement implements a partition of Σ and why invariant volumes reproduce the Born weights across runs.

We adopt three conservative premises that guide the rest of the paper:

1. **Minimal realism:** The state-space point ω denotes a physical microstate of the system.
2. **Deterministic dynamics:** Evolution is Hamiltonian flow on Σ generated by the physical Hamiltonian associated with the isolated system.
3. **Outcome as revelation:** A measurement is a controlled coupling that implements a partition Ω_i of a prepared region Ω_0 on Σ, with the registered result given by the unique cell containing ω.

These premises do not alter the mathematics of standard quantum theory or geometric quantum mechanics. They specify what the objects already in use are about. That is why an interpretation is needed, and why it can be evaluated on clear technical grounds.

1.1 Historical Development of Geometric Quantum Mechanics

The geometric formulation of quantum theory emerged from attempts to express the Hilbert-space formalism in coordinate-free language. Physicists and mathematicians gradually recognised that the physical content of pure-state quantum mechanics depends only on rays in Hilbert space, that is, equivalence classes under nonzero complex rescaling. These rays form a smooth complex projective manifold. Over several decades, this observation developed from a structural reformulation into the mature framework now known as geometric quantum mechanics.

1970s: Foundational reformulation: The modern formulation begins with Kibble's 1979 paper , *Geometrization of Quantum Mechanics*. Kibble showed that the space of pure states carries a natural symplectic structure, so that expectation values act as Hamiltonian functions and Schrödinger evolution becomes Hamiltonian flow on the projective manifold $\mathbb{CP}^{n-1}$. This established a precise parallel between classical Hamiltonian mechanics on phase space and quantum dynamics on a symplectic manifold, with the crucial difference that the quantum state manifold also carries an intrinsic complex structure. The result was a fully geometric restatement of the standard pure-state formalism.

1980s–1990s: Refinement and generalisation: Subsequent work clarified and strengthened this picture. Anandan and Aharonov (1990) formalised the role of the Fubini–Study metric on the projective space of rays and related its line element to quantum evolution and the time-energy uncertainty relation. Ashtekar and Schilling 1997) then presented a systematic account of the geometric formulation, showing that the pure-state manifold carries canonical Kähler structure, namely a compatible symplectic form, complex structure, and Riemannian metric. In this formulation, Schrödinger evolution is exactly the Hamiltonian flow generated by the energy function on the state manifold. Their treatment provided a coordinate-independent and geometrically complete representation of standard pure-state quantum mechanics.

1980s–2000s: Geometric phases and physical significance: In parallel, geometric ideas acquired direct physical significance through the study of phase. Berry (1984) showed that cyclic adiabatic evolution produces a measurable phase that depends only on the path taken in parameter space. Aharonov and Anandan (1987) generalised this result to arbitrary cyclic evolution, demonstrating that the associated phase has a natural geometric interpretation on projective Hilbert space. These developments made clear that quantum geometry is not merely formal packaging, but is tied to experimentally observable structure.

2000s: Consolidation and applications: By the early 2000s, geometric quantum mechanics had become a consolidated framework. Brody and Hughston 2001 developed a general treatment of observables as smooth real-valued functions $f : \Sigma \to \mathbb{R}$ whose associated Hamiltonian vector fields generate the dynamics. Bengtsson and Życzkowski 2006 synthesised the subject in their monograph *Geometry of Quantum States*, emphasising the metric, symplectic, and complex structure of $\mathbb{CP}^{n-1}$ and its applications to entanglement and quantum information. Related developments extended geometric methods to mixed states, quantum control, and holonomic quantum computation.

Present status: Geometric quantum mechanics is now a well-defined research area linking differential geometry, quantum theory, and quantum information. It provides a mathematically rigorous formulation in which the standard pure-state formalism is expressed in geometric terms. However, it remains largely noncommittal about ontology. The projective manifold is usually treated as a mathematical representation of state space, not necessarily as a space of physically real configurations. The open question, and the one relevant to the present work, is whether this geometry is merely a convenient reformulation or whether it should be regarded as physically fundamental.

1.2 What Geometric Quantum Mechanics Established

Geometric Quantum Mechanics (GQM) reformulates the standard pure-state formalism within an explicitly geometric framework. Its purpose is not to replace quantum mechanics, but to clarify its mathematical content by showing that the kinematics and dynamics of pure states are already encoded in the intrinsic geometry of projective state space. In this way, GQM isolates those features of quantum theory that follow from symmetry, topology, and differential structure, independently of any particular interpretation or measurement postulate.

For a finite-dimensional Hilbert space $\mathcal{H} \cong \mathbb{C}^n$, the space of pure states is the complex projective manifold

$$\mathbb{CP}^{n-1},$$

whose points correspond to equivalence classes of nonzero vectors $|\psi\rangle \in \mathcal{H}$ under nonzero complex rescaling. This manifold carries three mutually compatible geometric structures:

1. **A symplectic form** ω_Σ, which equips the state space with Hamiltonian structure. Expectation values of observables define Hamiltonian functions, and Schrödinger evolution is represented as the corresponding Hamiltonian flow.
2. **A complex structure** J, induced from multiplication by i on Hilbert space and projected to the ray space. This structure is compatible with the symplectic form and the metric.
3. **The Fubini–Study metric** g_{FS}, which provides a natural notion of distance and angle between pure states.

Together, these structures make the pure-state manifold into a Kähler manifold,

$$(\Sigma, \omega_\Sigma, J, g_{\mathrm{FS}}),$$

which is the canonical geometric setting for finite-dimensional pure-state quantum mechanics. This structure is not imposed ad hoc. It follows from the inner-product geometry of Hilbert space and hence from the linear structure of the standard formalism itself. Kibble 1979 first exhibited the correspondence between Schrödinger evolution and Hamiltonian flow, and Ashtekar and Schilling 1997 showed systematically that this Kähler structure underlies the geometric formulation of finite-dimensional quantum theory.

In this formulation, a Hermitian operator $\hat{A}$ is represented by the real-valued function

$$f_A([\psi]) = \frac{\langle\psi|\hat{A}|\psi\rangle}{\langle\psi|\psi\rangle}$$

on Σ. The Hamiltonian vector field X_H associated with the Hamiltonian function f_H is defined by

$$\iota_{X_H}\omega_\Sigma = df_H.$$

Schrödinger evolution is then precisely the Hamiltonian flow

$$\frac{d\omega}{dt} = X_H(\omega),$$

where $\omega \in \Sigma$ denotes the evolving state-space point. Because the flow is symplectic, it preserves the induced volume form. Thus the geometric formulation reproduces the standard deterministic dynamics of pure states exactly, while expressing it in differential-geometric rather than operator-theoretic language.

A further important result is measure-theoretic. Since $\mathrm{SU}(n)$ acts transitively on $\mathbb{CP}^{n-1}$ by isometries, the manifold carries a unique $\mathrm{SU}(n)$-invariant volume form, namely the Fubini–Study measure μ_{FS}. This measure provides the natural integration rule over pure-state space and is fixed by symmetry alone. Its existence and uniqueness are geometric consequences of the homogeneous structure of projective space and require no interpretive assumptions.

GQM also established that many familiar quantum notions admit direct geometric expression. Brody and Hughston 2001 showed that expectation values, uncertainties, and commutator structure can be written in terms of the symplectic and metric tensors on Σ. Geometric phases arise from the curvature of the natural bundle structure over projective space: parallel transport around a closed loop yields observable holonomy, including the Berry phase and its nonadiabatic generalisations (Berry 1984; Aharonov and Anandan 1987). More recent work connected the same geometric framework to state distinguishability, fidelity, entanglement geometry, and quantum information measures (Bengtsson and Życzkowski 2006; Anza and Crutchfield 2022.

By this stage, several points were firmly established:

- The pure-state space of any finite-dimensional quantum system is a finite-volume Kähler manifold.
- Schrödinger evolution is Hamiltonian flow preserving the symplectic volume form.
- $\mathrm{SU}(n)$ symmetry uniquely fixes the invariant operational measure on projective state space.
- Geometric phase effects and related interference phenomena have direct physical significance.

GQM therefore provides a rigorous, elegant, and empirically grounded reformulation of pure-state quantum mechanics. However, it deliberately stops short of assigning ontological status to the manifold or to its points. The geometry is usually treated as a representation of the formalism, not as a space of physically real configurations. The central questions therefore remain open: what, if anything, do the points of Σ represent physically, and how are definite outcomes to be understood within this geometric picture? Constraint-Surface Dynamics begins at exactly this point, by taking the geometric structure seriously as physical and asking what follows from that commitment.

1.3 What Geometric Quantum Mechanics Does Not Address

Although geometric quantum mechanics provides a mathematically elegant and empirically grounded reformulation of pure-state quantum theory, it was not designed to be a complete physical interpretation. Its central achievement is structural clarification, not ontological commitment. This distinction is important: GQM captures the geometric content of the formalism with precision, but it leaves open exactly the questions that an interpretation is supposed to answer.

First, GQM does not assign physical reality to the state manifold or to its points. In the standard literature, $\mathbb{CP}^{n-1}$ is treated as the natural manifold of pure states, that is, as the ray space associated with Hilbert space. Its points represent equivalence classes of nonzero Hilbert vectors under complex rescaling. What these points correspond to physically, if anything, is not specified. The geometry organises the formalism, but does not by itself tell us what in the world is geometrically structured.

Accordingly, when a system evolves by Hamiltonian flow on Σ, GQM does not claim that a physically real microstate traverses that trajectory. It states only that the Schrödinger dynamics of pure states can be represented as geometric flow on projective space. Whether Σ is an abstract representational space or a space of physically real configurations is left open. This interpretive neutrality is part of the framework's design.

Second, GQM does not solve the measurement problem. It shows how unitary evolution can be written as Hamiltonian flow and how the projective manifold carries a natural invariant measure, but it does not explain how a single definite outcome arises in an individual run. In the geometric formulation, the state remains a point on projective state space, and measurement is not introduced as a physical partitioning process on that manifold. No general mechanism identifies how a concrete interaction between system and apparatus yields a unique realised outcome region, nor how outcome selection is implemented dynamically.

Third, GQM does not derive the Born rule. The geometric formalism reproduces the standard kinematics and dynamics of pure states, but the amplitude-squared rule remains part of the empirical input. The geometry supplies symplectic flow, metric structure, and invariant measure, but not by itself a fully interpreted probability law for single-run outcomes. In that sense, GQM explains how amplitudes evolve, but not why laboratory frequencies take the Born form.

Fourth, GQM does not provide a preferred-basis mechanism. Because the projective manifold is homogeneous under the action of $\mathrm{SU}(n)$, all orthonormal bases are geometrically equivalent at the kinematic level. This symmetry is mathematically natural, but it leaves unresolved how one concrete measurement context becomes physically realised in experiment. The formalism does not specify how a basis is selected by an actual apparatus, nor how context-dependent outcomes become definite at the level of physical events.

More broadly, GQM inherits the interpretive ambiguity of the quantum state itself. Even when one passes from vectors $|\psi\rangle$ to rays $[\psi]$, the central question remains: is the state a physical object, a codification of information, or merely a calculational tool? GQM sharpens the geometric structure of the formalism, but it does not answer that question. It gives the structure, but not the referent.

This neutrality is both a strength and a limitation. It is a strength because it allows GQM to remain compatible with a wide range of interpretations. It is a limitation because compatibility is not explanation. A reformulation that remains equally hospitable to Copenhagen, Everettian, Bohmian 1952a, QBist, and decoherence-based readings does not by itself determine what the geometry means physically.

The point can be summarised briefly:

- GQM is a geometric reformulation of the pure-state formalism, not by itself an interpretation.
- It does not assign physical ontology to the manifold Σ or to its points.
- It does not solve the measurement problem, because it does not define a geometric mechanism for single-outcome selection.
- It does not derive the Born rule as an interpreted probability law for outcomes.
- It does not explain how a physically realised measurement basis is selected from the geometrically equivalent possibilities.

These are not defects in the original programme. They are the result of deliberate scope. The aim of GQM has been to clarify how quantum mechanics can be expressed geometrically, not to explain why nature behaves that way. Constraint-Surface Dynamics begins at exactly this boundary. It takes the geometric structure isolated by GQM and asks what follows if that structure is interpreted not merely as formal machinery, but as the configuration manifold of a physically real system.

1.4 The Interpretational Gap

Geometric Quantum Mechanics achieves a remarkable synthesis of mathematics and experiment. It shows that the pure-state formalism of quantum theory, including states, observables, and dynamics, can be expressed as differential geometry on the projective manifold $\mathbb{CP}^{n-1}$. Yet for all its elegance, it remains silent on the physical meaning of that geometry. The formalism specifies what evolves and how, but not what exists or why definite outcomes occur. This is the interpretational gap: the difference between a mathematically complete reformulation and a physically complete theory.

Within GQM, the geometry of Σ is fixed by the inner-product structure of Hilbert space. The Fubini–Study metric provides a natural notion of distance between pure states and is related to statistical distinguishability. The symplectic form supplies Hamiltonian flow and guarantees preservation of the induced volume form under unitary evolution. The complex structure renders these ingredients mutually compatible, making the state manifold Kähler and allowing expectation values, variances, and transition amplitudes to be expressed geometrically. These structures follow from symmetry and can be linked to experimentally accessible phenomena such as geometric phases and holonomies. In this sense, GQM establishes that pure-state quantum mechanics admits a fully geometric formulation with complete empirical adequacy.

Despite this success, GQM does not explain why any of these geometric objects should correspond to elements of physical reality. Several questions therefore remain open:

1. **What does the geometry represent?** Is Σ merely a convenient geometric representation of Hilbert-space rays, or is it a physically real configuration space in which systems possess definite microstates ω?
2. **Why do measurements yield single outcomes?** Hamiltonian flow on Σ is continuous and deterministic, whereas laboratory observations are discrete and definite. GQM does not supply a mechanism by which this geometry yields outcome selection.
3. **What is the status of the quantum state?** In geometric language, $|\psi\rangle$ is a coordinate representative of a ray $[\psi]$. But does that ray correspond to an ontic state of the system, an epistemic summary, or only a calculational device?
4. **Where do probabilities come from?** GQM contains the invariant Fubini–Study measure, but it does not by itself connect that measure to observed outcome frequencies. The Born rule remains an external input rather than a derived physical consequence.
5. **How does measurement act on the geometry?** The coupling of system and apparatus, the suppression of interference, and the registration of a definite result do not receive a direct geometric account in standard GQM.

These questions mark the boundary between mathematical formulation and physical interpretation. GQM remains on the mathematical side of that boundary.

This incompleteness is not a defect of the original programme. On the contrary, it is what makes the framework valuable. By making the formal structure of pure-state quantum theory explicit and self-consistent, GQM clarifies exactly where interpretation must begin. Once the projective manifold Σ is recognised as carrying the full kinematical structure of the theory, the natural next question is whether this geometry is merely useful or physically fundamental. If points $\omega \in \Sigma$ correspond to actual microstates evolving under deterministic Hamiltonian flow, then the probabilistic features of quantum mechanics may arise from geometric partitions and invariant volumes on Σ rather than from primitive indeterminism.

The interpretational gap therefore defines a concrete research direction:

- **Take the geometry seriously:** Treat Σ not merely as an abstract representation, but as the configuration manifold of an isolated physical system.
- **Define measurement geometrically:** Model observation as a physical interaction that induces a partition of Σ into disjoint outcome regions Ω_i.
- **Relate probability to symmetry:** Connect outcome weights to invariant volume ratios derived from the $\mathrm{SU}(n)$-fixed measure on Σ.
- **Preserve determinism:** Let the microstate $\omega(t)$ evolve continuously on Σ without collapse, while the registered outcome is determined by the unique outcome region occupied by ω.

These steps convert GQM from a purely mathematical reformulation into a candidate ontology capable of addressing definite outcomes and Born-form statistics within a single geometric framework. This ontological extension is precisely what Constraint-Surface Dynamics is intended to provide.

GQM shows that the machinery of quantum mechanics is already geometric. CSD asks whether that geometry is also physically real. The interpretational gap is therefore not a failure of GQM, but an invitation: to move from description to explanation, and from formal structure to physical meaning. The remainder of this paper takes up that invitation.

2 An Ontological Development of Geometric Quantum Mechanics

Geometric Quantum Mechanics provides a complete geometric reformulation of the pure-state formalism, but remains intentionally neutral about what that geometry represents physically. Constraint-Surface Dynamics begins at this point of neutrality and asks what follows if the relevant geometric structure is interpreted ontologically rather than merely formally. In CSD, the underlying manifold Σ is taken to be the configuration manifold of an isolated physical system. Each point $\omega \in \Sigma$ represents a definite microstate, and deterministic Hamiltonian flow on Σ represents the physical evolution of that system. Probabilities and outcome statistics are then attributed to invariant geometric structure on preparation and outcome regions, rather than to additional stochastic postulates.

2.1 From Formal Geometry to Physical Reality

Constraint-Surface Dynamics posits a finite-dimensional constraint surface Σ as the ontic state space. The manifold Σ is assumed to be smooth, compact, and equipped with symplectic structure

$$\omega_\Sigma \in \Omega^2(\Sigma).$$

The physical state of the system at time t is represented by a single point

$$\omega(t) \in \Sigma,$$

whose evolution is generated by a Hamiltonian function

$$H : \Sigma \to \mathbb{R}.$$

The associated Hamiltonian flow

$$\phi_t : \Sigma \to \Sigma$$

preserves the Liouville measure μ_Σ induced by the symplectic structure. Thus, for every measurable set $A \subset \Sigma$,

$$\mu_\Sigma(\phi_t(A)) = \mu_\Sigma(A).$$

This invariance is the basic deterministic measure-preserving structure on which the later probability discussion rests.

The complex projective manifold $\mathbb{CP}^{n-1}$ is not identified with Σ. Instead, it serves as an epistemic image of the ontic manifold. This image is represented through a smooth projection

$$\pi : \Sigma \longrightarrow \mathbb{CP}^{n-1},$$

which is generally many-to-one. The role of π is to encode coarse-graining: distinct ontic microstates on Σ may correspond to the same projective description in the quantum-effective sector. The conventional wavefunction $|\psi\rangle$ is therefore understood not as the ontic state itself, but as a coordinate representative of the projected description associated with $\pi(\omega)$, or more generally with a preparation region on Σ.

At the epistemic level, projective state space carries the Fubini–Study metric and its associated invariant measure μ_{FS}. In the quantum-effective sector, the operational measure on projective space is required to agree with the pushforward of the ontic measure under π, up to the normalisation fixed by the finite-volume structure. Thus for measurable sets $B \subset \mathbb{CP}^{n-1}$ one imposes the compatibility condition

$$\pi_*\mu_\Sigma(B) = \mu_\Sigma\big(\pi^{-1}(B)\big) \propto \mu_{\mathrm{FS}}(B),$$

with the proportionality fixed by normalisation on the relevant preparation domain. In this way, the deterministic measure-preserving dynamics on Σ induces the standard operational geometry on projective state space. The $\mathrm{SU}(n)$ invariance of μ_{FS} is then interpreted as the epistemic reflection of the underlying invariant structure on Σ.

Under this interpretation, the mathematical apparatus of geometric quantum mechanics describes the projected image of the deeper ontic dynamics rather than the ontic dynamics itself. The projective manifold and its associated homogeneous coordinates provide an epistemic description of the system at the level relevant for standard quantum predictions. The underlying physical evolution, however, is carried by the trajectory $\omega(t)$ on Σ. Observables, preparation domains, and outcome regions are represented fundamentally by measurable subsets of Σ, with their projective counterparts arising only after application of the projection map π.

This distinction separates ontic dynamics from epistemic representation. The evolution is deterministic on Σ, while the wavefunction formalism arises as a coarse-grained description on $\mathbb{CP}^{n-1}$. On this view, projective geometry is not discarded. It is retained, but reinterpreted as the geometric image of a deeper measure-preserving dynamics. The structure of $\mathbb{CP}^{n-1}$ therefore reflects the operational geometry induced from Σ, rather than constituting the fundamental state space itself.

2.2 Key Additions to Geometric Quantum Mechanics

Constraint-Surface Dynamics extends geometric quantum mechanics by introducing an explicit distinction between the ontic structure of the theory and its epistemic representation. The purpose of this extension is not to alter the successful formal content of quantum theory, but to specify what that formalism is taken to describe physically. The resulting framework links deterministic dynamics on the constraint surface to the standard projective formalism through a small number of structural principles.

1. Ontological layer: The manifold Σ is taken to be physically real. Each isolated system occupies a single microstate

$$\omega \in \Sigma,$$

and its evolution is given by deterministic Hamiltonian flow generated by a Hamiltonian

$$H : \Sigma \to \mathbb{R}$$

with symplectic form ω_Σ. The associated flow

$$\phi_t : \Sigma \to \Sigma$$

preserves the Liouville measure μ_Σ. All physical histories are therefore curves $\omega(t)$ on Σ. On this view, the ontic content of the theory lies in the evolving microstate and the invariant geometric structure of the manifold.

2. Measurement as partition: A measurement is represented by a finite measurable partition of a prepared region $\Omega_0 \subset \Sigma$,

$$\Omega_0 = \bigsqcup_i \Omega_i.$$

The observed outcome is determined by the unique cell Ω_i containing the actual microstate at the relevant time. No physical discontinuity or stochastic collapse is introduced at the ontic level. What changes is the epistemic description associated with the projected image of the post-interaction state under

$$\pi : \Sigma \to \mathbb{CP}^{n-1}.$$

Thus measurement is treated as physical partitioning of the preparation domain, not as a separate non-Hamiltonian law.

3. Isolation principle: Quantum-effective behaviour is associated with physical isolation. For a sufficiently decoupled system, the relevant ontic dynamics are described on a subsystem manifold Σ_{sys}. When the system interacts with an environment or apparatus, the joint dynamics are defined on a larger manifold describing the coupled degrees of freedom. In the simplest idealised case this may be represented by a product structure,

$$\Sigma_{\text{sys+env}} = \Sigma_{\text{sys}} \times \Sigma_{\text{env}},$$

though more generally the coupled ontic structure need not remain globally factorised. The essential point is that interaction modifies the effective partition structure relevant to the subsystem description. Decoherence is interpreted as loss of isolation: overlapping ontic configurations in the isolated description are mapped, under interaction, into effectively disjoint epistemic descriptions for the measured outcomes.

4. Born-form weights from geometric symmetry: The group $\mathrm{SU}(n)$ acts transitively and isometrically on $\mathbb{CP}^{n-1}$. In the quantum-effective sector, the pushforward of the ontic invariant measure induces the standard invariant operational measure on projective state space. For a measurement partition $\{\Omega_i\}$ of the preparation region Ω_0, the natural geometric weight is

$$P(i) = \frac{\mu_\Sigma(\Omega_i)}{\mu_\Sigma(\Omega_0)}.$$

These weights agree with Born-form probabilities only after the standard operational consistency conditions are imposed on the admissible probability functional on $\mathbb{CP}^{n-1}$, together with the typicality-to-frequency link for repeated trials. In this sense, CSD does not append the Born rule as an independent stochastic axiom. Rather, it treats Born-form weights as the operational image of invariant geometric structure in the quantum-effective regime.

5. Finiteness and compactness: The ontic manifold is assumed to satisfy

$$0 < \mu_\Sigma(\Sigma) < \infty,$$

so that the relevant preparation and outcome measures are finite and normalisable. Compactness guarantees that the total invariant measure is well defined and conserved under Hamiltonian evolution. This finiteness assumption is central to the present formulation: it supports normalised volume ratios, typicality arguments for long-run frequencies, and the finite-dimensional scope of the theory.

Taken together, these additions convert GQM from a purely structural reformulation into an ontological framework. The mathematical objects of the geometric formalism are retained, but they are now assigned distinct physical roles: Σ is ontic, $\mathbb{CP}^{n-1}$ is epistemic, measurement is partition, and outcome weights arise from invariant geometry in the appropriate operational sector.

6. Epistemic and ontic distinction: CSD maintains a strict separation between underlying ontology and operational representation. The manifold Σ, its symplectic structure ω_Σ, and its invariant measure μ_Σ belong to the ontic layer. By contrast, projective state space $\mathbb{CP}^{n-1}$, wavefunction representatives $|\psi\rangle$, and the Fubini–Study measure μ_{FS} belong to the epistemic layer associated with the projection map π. This distinction allows the framework to preserve the standard operational predictions of quantum theory while assigning clear physical meaning to the deeper geometric structure from which those predictions arise.

2.3 The Projection Map and the Structure of Epistemic Coordinates

The manifold Σ represents the ontic state space in which the microstate ω evolves deterministically. In general, observers do not have direct access to ω. What is operationally accessible is the projected state

$$\pi(\omega) \in \mathbb{CP}^{n-1},$$

which represents a coarse-grained description of the underlying ontic configuration. Distinct points of Σ may therefore map to the same projective ray $[\psi]$, reflecting limited operational resolution rather than multiplicity of physical reality. In this sense, $\mathbb{CP}^{n-1}$ is epistemic: it records the maximally refined projective description available in the quantum-effective sector, not the underlying ontic microstate itself.

For a preparation region $\Omega_0 \subset \Sigma$, the associated epistemic description is the image

$$\pi(\Omega_0) \subset \mathbb{CP}^{n-1}.$$

A particular choice of homogeneous coordinates on this image yields an ordinary wavefunction representative $|\psi\rangle$. The wavefunction is therefore not itself an ontic variable in CSD. It is a coordinate description of the projected preparation class determined by π.

The projection map is required to be compatible with measure in the quantum-effective sector. Let μ_Σ denote the Liouville measure on Σ. The induced measure on projective space is the pushforward measure

$$\pi_*\mu_\Sigma,$$

defined for measurable sets $C \subset \mathbb{CP}^{n-1}$ by

$$\pi_*\mu_\Sigma(C) = \mu_\Sigma\big(\pi^{-1}(C)\big)\,.$$

Within the quantum-effective sector, this induced operational measure is identified, up to the normalisation fixed by finite total volume, with the Fubini–Study measure μ_{FS}. Thus the standard projective measure is not introduced as an independent stochastic ingredient. It is the operational image of the invariant ontic measure under π.

The action of $\mathrm{SU}(n)$ on $\mathbb{CP}^{n-1}$ is interpreted as the induced action of symmetries on the epistemic layer. More precisely, for each

$$g \in \mathrm{SU}(n),$$

one requires a corresponding transformation Φ_g on the relevant sector of Σ such that

$$\pi \circ \Phi_g = g \circ \pi.$$

This intertwining condition ensures that invariance of the operational measure on projective space reflects invariance of the underlying ontic measure under the corresponding canonical symmetries.

Measurement partitions are defined fundamentally on Σ. If a prepared region admits a measurable partition

$$\Omega_0 = \bigsqcup_i \Omega_i,$$

then the corresponding epistemic images are

$$C_i = \pi(\Omega_i) \subset \mathbb{CP}^{n-1}.$$

The associated operational weight is inherited from the ontic geometry:

$$P(i) = \frac{\mu_\Sigma(\Omega_i)}{\mu_\Sigma(\Omega_0)}.$$

Equivalently, in the quantum-effective sector and after normalisation of the induced projective measure,

$$P(i) = \frac{\mu_{\mathrm{FS}}(C_i)}{\mu_{\mathrm{FS}}(C_0)}.$$

These quantities agree with Born-form weights only after the admissible probability functional is fixed by the standard operational consistency conditions discussed elsewhere in the paper. The appearance of Hilbert-space amplitudes is therefore understood as a feature of the epistemic coordinate description induced by π, not as an independent ontological postulate.

On this view, the complex projective manifold is the epistemic shadow of the ontic dynamics on Σ. Its metric, symmetry, and invariant measure do not replace the underlying deterministic geometry. Rather, they arise as the operationally accessible image of that geometry in the quantum-effective regime. Quantum mechanics then appears as a representational theory built on a deeper deterministic dynamics, with $\mathbb{CP}^{n-1}$ describing what can be inferred about the ontic evolution on Σ rather than what exists fundamentally.

2.4 Resolving the GQM Gaps

Constraint-Surface Dynamics is intended to complete the geometric programme at precisely the points where standard GQM remains interpretively neutral. Its central claims may be summarised as follows:

- **Ontology:** The manifold Σ is taken to be physically real, and each isolated system occupies a definite microstate $\omega \in \Sigma$.
- **Measurement:** Outcomes are associated with measurable regions Ω_i of a prepared domain $\Omega_0 \subset \Sigma$, with the realised result determined by the unique region containing the actual microstate.
- **Probability:** Operational outcome weights are grounded in invariant geometric volume ratios on Σ, with Born-form probabilities recovered in the quantum-effective sector through the induced projective measure together with the admissible probability functional.
- **Measurement context:** The symmetry of projective space does not by itself select a preferred basis. In CSD, basis selection is replaced by context-defined partition structure induced by the physical measurement interaction.
- **Observer role:** Observers and apparatus are treated as physical systems embedded in the same dynamical framework, so outcome registration is not primitive but secondary to the underlying interaction and partition structure.
- **Finiteness:** Finite total invariant measure ensures normalisability, conservation of outcome weights under isolated evolution, and the stability of long-run statistical description.

2.5 Why This Extension Is Natural

Constraint-Surface Dynamics introduces no new stochastic law, no collapse postulate, and no additional empirical constants. Its central move is interpretive rather than algebraic: it takes the geometric structure already isolated by GQM and asks what follows if that structure is understood as physically instantiated. Once pure-state quantum evolution is recognised as Hamiltonian flow on a finite-dimensional Kähler manifold, it is natural to ask whether the underlying geometry should be treated as ontic rather than merely representational, and whether the standard projective formalism should be understood as the epistemic image of that deeper structure.

On this view, the extension is economical. The dynamical framework is retained, the standard operational predictions are preserved, and the additional explanatory work is carried by the ontic interpretation of Σ, the partition-based account of measurement, and the geometric account of outcome weights. The resulting picture is deterministic, finite in scope, and compatible with the standard quantum formalism in the regime treated in this paper.

CSD is therefore proposed as a realist completion of GQM in which the mathematics remains unchanged at the operational level, while its physical meaning is made explicit. In that sense, the framework is intended as a minimal ontological development of geometric quantum mechanics: the same geometric apparatus, but with a definite account of what exists, how measurement is represented, and how Born-form statistics arise in the quantum-effective sector.

3 What Would Constitute a Solution

A viable completion of quantum mechanics must do more than reproduce experimental data. It must explain, in clear physical terms, how definite outcomes and stable statistical regularities arise within a deterministic dynamical framework while preserving the empirically confirmed structure of standard quantum theory. The following criteria define what would count as such a solution and provide the benchmark against which Constraint-Surface Dynamics is assessed.

3.1 Definite Outcomes

Any successful account must explain why each experimental run yields one definite result. Laboratory detectors do not record superposed outcomes, and observers agree on the event that is registered. Linear evolution alone does not by itself explain this empirical definiteness.

In CSD, the issue is addressed at the ontic level. At every time, the system occupies a single microstate

$$\omega \in \Sigma.$$

During a measurement interaction, the relevant preparation region Ω_0 is partitioned into measurable outcome regions $\{\Omega_i\}$. The realised outcome is determined by the unique region containing the actual microstate. No physical collapse is introduced. What changes is the observer's epistemic description of which outcome region contains ω. On this view, definite outcomes follow from the definiteness of the ontic state together with the partition structure induced by the measurement context.

3.2 Origin of the Born Rule

A complete theory should also explain why observed frequencies take the Born form. In CSD, this is not treated as a primitive stochastic postulate. Instead, the theory proposes a modular chain of explanation. First, invariant geometry on Σ determines natural volume ratios on preparation and outcome regions. Second, in the quantum-effective sector, the induced operational measure on projective space is fixed by symmetry. Third, the admissible probability functional is constrained by the standard operational consistency conditions. Finally, typicality links these weights to observed long-run frequencies.

Thus the geometric quantity

$$P(i) = \frac{\mu_\Sigma(\Omega_i)}{\mu_\Sigma(\Omega_0)}$$

provides the ontic weight associated with outcome region Ω_i, while Born-form probabilities arise at the operational level once the projective representation is imposed. In this way, CSD seeks to replace a primitive probability axiom with a geometric account of outcome weights together with a typicality-based frequency interpretation.

3.3 Observer Neutrality

An interpretation should not assign a special dynamical status to observers. Measurement must be described as an ordinary physical interaction governed by the same general principles as the rest of the theory.

In CSD, observers and measuring devices are treated as physical subsystems within the same overall framework. Registration of an outcome corresponds to the formation of stable correlations between subsystem degrees of freedom and the ontic microstate. The observer is therefore not a primitive ingredient in the dynamics. Observation is secondary to the interaction-defined partition structure and the resulting correlation between system, apparatus, and recorded outcome.

3.4 Empirical Equivalence with Quantum Mechanics

Any viable ontological completion must reproduce the empirically confirmed predictions of standard quantum mechanics in the domain where those predictions are known to hold. In particular, it must preserve the observed quantitative structure of interference, transition amplitudes, and experimentally accessible correlations.

CSD is intended to preserve the standard operational structure in the finite-dimensional quantum-effective sector. The projective description retains the usual geometric representation of unitary evolution, and the operational probabilities are required to match the standard Born-form expressions in that regime. The claim is therefore not that CSD changes the successful predictions of quantum theory, but that it reinterprets the underlying geometric structure from which those predictions arise.

3.5 Measurement Context and Basis Selection

A satisfactory completion should also explain how concrete measurement contexts arise without introducing an unexplained privileged basis into the underlying kinematics. Standard quantum theory is invariant under unitary transformations, and any completion should preserve that symmetry at the fundamental level.

In CSD, no basis is kinematically preferred. The symmetry of projective state space is retained. What is physically selected in a given experiment is not a globally preferred basis, but a context-defined partition induced by the specific interaction Hamiltonian of the measurement arrangement. Apparent basis selection is therefore attributed to the concrete measurement context rather than to any privileged direction built into the state space itself.

3.6 Conceptual and Mathematical Coherence

Beyond predictive adequacy, the framework must be internally coherent. At minimum this requires:

- a mathematically well-defined deterministic flow on the ontic manifold,
- conservation of the invariant measure under isolated evolution,
- stability of the relevant outcome weights under the stated dynamical assumptions,
- and compatibility with the operational causal constraints respected by standard quantum theory in the regime considered.

CSD is designed to satisfy these requirements within its present finite-dimensional scope. Deterministic Hamiltonian flow provides continuous ontic evolution, finite invariant measure ensures normalisability of preparation and outcome weights, and the same geometric structure that governs motion also underlies the statistical description. The intended result is a single framework in which deterministic dynamics and Born-form outcome statistics are connected rather than postulated independently.

3.7 Minimality and Parsimony

A convincing completion should introduce no unnecessary dynamical laws, free parameters, or stochastic mechanisms. On its own terms, CSD is intended as a minimal realist development of geometric quantum mechanics. It does not modify the operational equations of the standard finite-dimensional formalism, does not introduce collapse dynamics, and does not add new empirical constants. Its central additional commitment is ontological: it treats the underlying geometric manifold as physically real and interprets invariant measure structure as the basis of outcome weights. The hope is that, from this single shift in interpretation, definite outcomes, Born-form statistics, and the measurement process can be understood within one unified geometric picture.

4 CSD in a Nutshell

4.1 Building on Geometric Quantum Mechanics

Geometric quantum mechanics provides the formal framework on which the present construction is built. In the standard geometric formulation, the pure states of an n-level quantum system are represented by points of the complex projective manifold

$$\mathbb{CP}^{n-1},$$

which carries a canonical Kähler structure. Its symplectic form ω_{FS}, compatible complex structure J, and Fubini–Study metric g_{FS} together equip $\mathbb{CP}^{n-1}$ with the structure required for Hamiltonian dynamics on pure-state space. The unitary symmetry group $\mathrm{SU}(n)$ acts transitively and isometrically on this manifold and fixes the unique invariant operational measure μ_{FS}.

Within this framework, Schrödinger evolution is represented as Hamiltonian flow on projective state space. Given a Hermitian Hamiltonian operator $\hat{H}$, the associated real Hamiltonian function on $\mathbb{CP}^{n-1}$ is

$$H([\psi]) = \frac{\langle\psi|\hat{H}|\psi\rangle}{\langle\psi|\psi\rangle}.$$

The corresponding Hamiltonian vector field X_H is defined by

$$\iota_{X_H}\omega_{\mathrm{FS}} = dH, \qquad \dot{\gamma}(t) = X_H(\gamma(t)).$$

Liouville's theorem then guarantees preservation of the invariant volume form under the induced flow. In this sense, standard pure-state quantum mechanics already appears as deterministic Hamiltonian evolution on a compact Kähler manifold.

Constraint-Surface Dynamics retains this mathematical structure at the operational level, but interprets it differently. The ontic state space is not $\mathbb{CP}^{n-1}$ itself, but a finite-dimensional symplectic manifold Σ equipped with Liouville measure μ_Σ and Hamiltonian flow ϕ_t. The projective manifold arises as an epistemic image through a smooth surjective map

$$\pi : \Sigma \to \mathbb{CP}^{n-1}.$$

The conventional wavefunction $|\psi\rangle$ is then understood as a coordinate representative of the projected description $\pi(\omega)$, where

$$\omega \in \Sigma$$

is the actual ontic microstate of the system. Distinct ontic microstates may project to the same ray $[\psi]$, reflecting coarse-graining at the epistemic level rather than physical multiplicity at the ontic level.

In the quantum-effective sector, the projected dynamics is required to reproduce standard Schrödinger evolution on $\mathbb{CP}^{n-1}$. Thus, if the ontic dynamics on Σ are generated by an appropriate Hamiltonian H_Σ, then the induced evolution of $\pi(\omega(t))$ matches the usual unitary flow on projective state space. No modification of standard quantum dynamics is introduced at the operational level. The difference lies in interpretation: the geometry of Σ is taken to be physically real, whereas the Hilbert-space and projective descriptions are understood as epistemic representations of that deeper geometry.

Because $\mathbb{CP}^{n-1}$ is compact, the induced operational measure has finite total volume. In the quantum-effective sector, the pushforward of the ontic measure under π is identified, after normalisation, with the Fubini–Study measure:

$$\pi_*\mu_\Sigma \propto \mu_{\mathrm{FS}}.$$

Equivalently, for measurable sets $C \subset \mathbb{CP}^{n-1}$,

$$\pi_*\mu_\Sigma(C) = \mu_\Sigma\big(\pi^{-1}(C)\big)\,,$$

with the proportionality constant fixed by normalisation on the relevant finite-volume domain. Since μ_Σ is preserved by Hamiltonian flow on Σ, the induced operational measure is conserved under the projected dynamics. This ensures normalisable probabilities and stable ensemble weights in the epistemic layer without the need for external cut-offs. The invariance of μ_{FS} under $\mathrm{SU}(n)$ is therefore interpreted as the operational image of the corresponding invariant structure on the ontic manifold.

4.2 The Core Ontological Claim

Constraint-Surface Dynamics asserts that the configuration surface Σ is the fundamental ontic space in which physical systems evolve. Each point

$$\omega \in \Sigma$$

specifies a complete micro-configuration of the degrees of freedom permitted by the constraints defining the system. The physical state is therefore represented by a single trajectory

$$\omega(t) \in \Sigma,$$

generated by a Hamiltonian flow

$$\phi_t : \Sigma \to \Sigma$$

that preserves both the symplectic structure ω_Σ and the Liouville measure μ_Σ. No probabilistic superposition is introduced at the ontic level. All indeterminacy enters through epistemic coarse-graining via the projection map π discussed above.

Observable statistics arise from the way the conserved flow acts on measurable regions of the ontic surface. When a system interacts with an apparatus or environment, the interaction defines a measurement context and induces a measurable partition of the prepared region

$$\Omega_0 = \bigsqcup_i \Omega_i \subset \Sigma.$$

Because the ontic dynamics preserve the invariant measure, the corresponding geometric weights

$$\frac{\mu_\Sigma(\Omega_i)}{\mu_\Sigma(\Omega_0)}$$

are stable under the isolated evolution relevant to the given context. In the CSD framework, these ratios provide the ontic basis for outcome weights. Their connection to observed long-run frequencies is supplied by the typicality framework, while their connection to Born-form probabilities in the projective description is supplied, in the quantum-effective sector, by the induced operational measure together with the admissible probability functional.

In the present formulation, Σ is taken to be a finite-dimensional symplectic manifold that encodes the ontic content of the isolated system. Correspondence with spacetime coordinates is not fundamental at this level, but arises only after further coarse-graining and effective description. The projection map π induces epistemic states on $\mathbb{CP}^{n-1}$, and additional projections may be introduced to define effective macroscopic coordinates relevant to observers and apparatus. The detailed emergence of Lorentzian spacetime structure, including causal cones and curvature, is not developed in this paper and is deferred to later work.

The ontological commitments of the present framework may therefore be stated as follows:

1. A physical system follows a single deterministic trajectory $\omega(t)$ on Σ.
2. The Liouville measure μ_Σ is invariant under the Hamiltonian flow ϕ_t, ensuring conservation of total measure and stability of the corresponding geometric outcome weights.
3. Apparent randomness arises from epistemic restriction to a context-defined outcome region rather than from ontic indeterminism.
4. Spacetime relations and relativistic structure are treated as emergent and are not derived in the present paper.

This formulation grounds measurement and quantum statistics in a deterministic geometric framework while keeping the present scope modest. It clarifies both what CSD claims here and what remains open for later stages of the programme.

4.3 Dynamics and Hamiltonian Flow

For an isolated system in the quantum-effective sector, the operational state manifold is $\mathbb{CP}^{n-1}$. The ontic manifold is Σ, equipped with a surjective projection

$$\pi : \Sigma \to \mathbb{CP}^{n-1},$$

which realises the projective description as an epistemic representation of the underlying ontic dynamics. The projective manifold carries the standard geometric structures of GQM, namely the Fubini–Study symplectic form, the compatible complex structure J, and the Fubini–Study metric g_{FS}. These determine the unique $\mathrm{SU}(n)$-invariant operational measure on $\mathbb{CP}^{n-1}$. At the ontic level, Σ carries its own symplectic form ω_Σ and the associated Liouville measure μ_Σ, which is preserved under Hamiltonian flow.

The ontic Hamiltonian

$$H_\Sigma : \Sigma \to \mathbb{R}$$

generates deterministic motion through the Hamiltonian vector field X_{H_Σ} defined by

$$\iota_{X_{H_\Sigma}} \omega_\Sigma = dH_\Sigma.$$

The equations of motion are

$$\dot{\omega}(t) = X_{H_\Sigma}(\omega(t)).$$

Because the flow is symplectic, it preserves the Liouville measure:

$$\mathcal{L}_{X_{H_\Sigma}} \mu_\Sigma = 0.$$

The ontic dynamics therefore define deterministic, measure-preserving evolution on the finite manifold Σ.

CSD differs from standard GQM not at the level of operational mathematics, but at the level of interpretation. The trajectory $\omega(t)$ is taken to represent the actual ontic history of the system, rather than merely a formal representative of a quantum state. The invariant measure on Σ is correspondingly interpreted as physically relevant geometric structure, with its conservation providing the basis for stable outcome weights under isolated evolution.

Because Σ is assumed compact within the present formulation, the flow is complete on the relevant domain, normalised invariant measure exists, and the theory admits finite preparation and outcome weights. Conserved quantities arise, as usual, from continuous symmetries of the Hamiltonian.

Quantum-effective behaviour is associated with dynamical isolation. An isolated subsystem evolves on its own ontic manifold Σ_{sys}, governed by an effective Hamiltonian H_{sys}. When the subsystem interacts with an apparatus or environment, the relevant ontic description must be enlarged to include the coupled degrees of freedom. In an idealised setting this may be represented by

$$\Sigma_{\mathrm{tot}} = \Sigma_{\mathrm{sys}} \times \Sigma_{\mathrm{env}},$$

together with an interaction term in the total Hamiltonian. More generally, the coupled ontic structure need not remain globally factorised, but the essential point is that the interaction changes the effective partition structure relevant to the subsystem description. Decoherence is then interpreted as loss of isolation: information that was effectively confined within the isolated subsystem description is dispersed into the larger coupled dynamics, suppressing the operational accessibility of interference.

Within this geometric ontology, a measurement is not a stochastic collapse but a context-defining interaction that induces a measurable partition of the prepared region $\Omega_0 \subset \Sigma$ into disjoint outcome regions Ω_i. The system's ontic microstate $\omega(t)$ always lies in exactly one such region, and the observed outcome is the registration of that fact at the epistemic level. The corresponding ontic weight is

$$P(i) = \frac{\mu_\Sigma(\Omega_i)}{\mu_\Sigma(\Omega_0)}.$$

Because the ontic flow preserves μ_Σ, these geometric weights are stable under the isolated evolution relevant to the measurement context. In the quantum-effective sector, after projection to $\mathbb{CP}^{n-1}$ and imposition of the standard operational consistency conditions, these weights recover the Born-form probabilities. Measurement is therefore interpreted as revelation of which region of Σ contains the actual microstate, not as a fundamental discontinuity in the dynamics.

4.4 How Measurement Works

Measurement is treated as a physical process within the same deterministic framework that governs all other evolution. No separate collapse law is introduced. A measurement interaction is a temporary coupling that alters the effective dynamical context of the system and induces a partition of the relevant preparation region. What appears as collapse in the standard formalism is, on this view, an epistemic update: a change in what embedded observers can infer about the location of the ontic microstate ω.

At all times the system occupies a single definite point

$$\omega \in \Sigma.$$

The manifold Σ is the ontic configuration space, and Hamiltonian flow carries $\omega(t)$ continuously along its trajectory. There is therefore no ontic superposition in the sense of multiple simultaneously realised outcomes. The superposed amplitudes that appear in a Hilbert-space description belong to the epistemic layer and reflect the projected representation of incomplete information about the underlying ontic state.

When a system is dynamically isolated, its trajectory evolves on the corresponding subsystem manifold Σ_{sys} under its effective Hamiltonian. A measurement begins when that isolation is broken through coupling to an apparatus and, typically, to an environment. In an idealised description the relevant ontic manifold is enlarged to

$$\Sigma_{\text{tot}} = \Sigma_{\text{sys}} \times \Sigma_{\text{app}} \times \Sigma_{\text{env}},$$

with total Hamiltonian

$$H_{\text{tot}} = H_{\text{sys}} + H_{\text{app}} + H_{\text{env}} + H_{\text{int}}.$$

More generally, the coupled ontic structure need not remain globally factorised, but the essential point is that the interaction defines a new dynamical context on the enlarged manifold. This coupling phase is the loss of isolation relevant to measurement. Quantum-effective interference requires isolation; classical definiteness arises when the subsystem becomes stably correlated with apparatus and environment degrees of freedom.

The interaction induces a finite measurable partition of the prepared region

$$\Omega_0 \subset \Sigma_{\text{tot}}$$

into disjoint outcome regions

$$\Omega_0 = \bigsqcup_i \Omega_i.$$

Each region corresponds to a distinct stable macroscopic registration configuration of the apparatus. The ontic microstate always lies in exactly one such region, and the measurement outcome is the epistemic identification of that region. Nothing discontinuous happens to the ontic state itself.

The corresponding ontic outcome weight is

$$P(i) = \frac{\mu_\Sigma(\Omega_i)}{\mu_\Sigma(\Omega_0)}.$$

Because the relevant ontic flow preserves the invariant measure, these geometric weights are stable under the isolated evolution associated with the measurement context. In the quantum-effective sector, after projection to $\mathbb{CP}^{n-1}$ and application of the admissible operational probability functional, these weights reproduce Born-form probabilities. Measurement statistics are therefore grounded in invariant geometric structure rather than in primitive stochastic collapse.

Observers and apparatus are not external to the dynamics. Their internal degrees of freedom are part of the same total ontic evolution. Observation is thus a process of correlation within Σ_{tot}: the observer's recorded state becomes correlated with the outcome region that already contains the system's ontic microstate. Agreement between observers follows from the shared dynamical and geometric structure of the coupled system, not from any privileged external standpoint.

Once an outcome is registered, the observer updates the relevant epistemic description from the initial preparation region Ω_0 to the revealed outcome region Ω_i. This update changes knowledge, not ontology. The microstate ω has not jumped. It has followed a continuous trajectory throughout. The orthodox collapse rule is therefore replaced by conditioning on the realised outcome region within the invariant measure structure of the ontic manifold.

On this account, measurement is a deterministic interaction that destroys subsystem isolation and yields a definite registered result through context-defined partitioning. The system is never ontically superposed, never undergoes physical collapse, and never leaves the deterministic manifold Σ. Observers, apparatus, and outcomes all belong to one continuous Hamiltonian dynamics, with operational probabilities arising from invariant geometric weights.

4.5 Relationship to Standard Quantum Mechanics

Constraint-Surface Dynamics is intended to preserve the empirically confirmed predictions of standard quantum mechanics within its present finite-dimensional scope while changing the underlying ontology. At the operational level, it retains the full geometric structure used in GQM: projective state space, the Fubini–Study metric, the symplectic description of unitary evolution, and the invariant operational measure. What changes is the interpretation of how this operational structure arises. In CSD, the projective formalism is not fundamental. It is the epistemic image of a deeper deterministic dynamics on the ontic manifold Σ.

For any experiment describable within the standard finite-dimensional formalism, the aim is that CSD yields the same numerical predictions. Expectation values, transition weights, interference patterns, and correlation structure are preserved because the projective description induced from Σ is required to reproduce the usual geometric dynamics on $\mathbb{CP}^{n-1}$ in the quantum-effective sector. Outcome weights are assigned geometrically at the ontic level through invariant volume ratios on preparation and outcome regions, and these agree operationally with Born-form probabilities once the admissible projective probability functional is imposed. In this sense, CSD is not introduced as an empirically distinct replacement for quantum mechanics in the regime treated here, but as an ontological completion of its geometric formulation.

The difference lies in what the mathematical objects are taken to represent. In standard quantum mechanics and in ordinary presentations of GQM, projective state space is a state manifold for the formalism, while the ontological status of the wavefunction or ray remains underdetermined. In CSD, the ontic manifold Σ is physically real. The deterministic trajectory

$$\omega(t) \in \Sigma$$

represents the actual microhistory of the system. The projective state, by contrast, belongs to the epistemic layer: it encodes only the coarse-grained information accessible through the projection map π.

On this view, the wavefunction $|\psi\rangle$ remains useful but loses fundamental ontic status. It becomes a coordinate representative of the projected epistemic description associated with a preparation region on Σ. Its amplitudes do not describe simultaneously realised ontic alternatives. Rather, they encode the operational geometry inherited from the underlying invariant measure structure. The amplitude-squared quantities that appear in the standard formalism are therefore interpreted as the projective image of underlying geometric weights on the ontic manifold.

The relation between standard ψ-based quantum theory and CSD may therefore be understood as analogous, in broad structural terms, to the relation between a macroscopic effective description and a deeper dynamical substrate. The operational formalism captures stable regularities in the projected description, while the ontic dynamics on Σ supply the deeper deterministic structure from which those regularities arise. The point of the analogy is not to identify quantum theory with classical statistical mechanics, but to emphasise that an effective probabilistic description can coexist with an underlying deterministic geometry.

By adding this ontological layer to GQM, CSD aims to provide a unified framework in which the mathematical structure of standard quantum theory is preserved while its physical meaning is made explicit. The manifold Σ is ontic, $\omega(t)$ is the actual trajectory, and the projective formalism is epistemic. On this picture, nothing changes in the empirically successful operational content of quantum mechanics within the present scope, but the interpretational gap between formalism and ontology is narrowed by assigning a definite physical role to the underlying geometry.

5 Mathematical Framework

5.1 The Configuration Manifold Σ

Following the geometric quantum mechanics framework developed by Kibble 1979 , Ashtekar and Schilling 1997, Brody and Hughston 2001, and Bengtsson and Życzkowski 2006, we distinguish between the projective state manifold and the deeper ontic manifold used in CSD. For a finite-dimensional n-level system, the operational pure-state space is

$$\mathbb{CP}^{n-1},$$

equipped with its standard Fubini–Study metric, compatible symplectic structure, and complex structure. In CSD, this projective manifold is interpreted as epistemic rather than ontic. The ontic state space is a finite-dimensional symplectic manifold Σ, together with a projection

$$\pi : \Sigma \to \mathbb{CP}^{n-1}.$$

What CSD adds is not new projective geometry, but a new interpretation: points

$$\omega \in \Sigma$$

are taken to represent actual physical microstates, not merely formal coordinates.

We assume throughout that Σ is a smooth symplectic manifold

$$(\Sigma, \omega_\Sigma),$$

with ω_Σ closed and non-degenerate. The projective manifold $\mathbb{CP}^{n-1}$ carries its standard Kähler structure, and in the quantum-effective sector the projection π is required to induce the standard operational geometry on projective state space. The corresponding invariant operational measure is the Fubini–Study measure μ_{FS}, uniquely fixed by the transitive isometric action of $\mathrm{SU}(n)$ on $\mathbb{CP}^{n-1}$.

Two distinct finiteness statements should be separated carefully.

Mathematical finiteness: The manifold $\mathbb{CP}^{n-1}$ is compact. Equipped with the Fubini–Study metric, it therefore has finite total volume. This is a purely geometric property of the finite-dimensional projective state space and requires no extra cutoff or additional scale.

Physical finiteness in CSD: CSD interprets this finite-volume geometry as physically significant for the accessible ontic state space of an isolated finite-dimensional system. At this stage, no numerical upper bound is introduced, and no identification is made with any specific Planck-scale density or cutoff. The present claim is more modest: the compact finite-dimensional geometry used in GQM is taken to correspond to a physically instantiated finite ontic domain in the regime considered here.

This separation matters. Compactness implies finite total measure mathematically. The physical interpretation of that finiteness is an additional ontological commitment, not a new geometric theorem.

Examples: The simplest case is a qubit. For $n = 2$, the operational state space is

$$\mathbb{CP}^1 \cong S^2,$$

the Bloch sphere. The induced Hamiltonian vector fields reproduce the standard geometric description of spin precession. For higher dimensions, $\mathbb{CP}^{n-1}$ remains compact and Kähler, with well-defined symplectic and metric structure and the same unique invariant operational measure. The projective geometry therefore continues to provide the finite-volume operational state space used in the standard geometric formulation.

In CSD, however, this projective geometry is not itself fundamental. The ontic manifold is Σ, and the projective description arises through π. The actual state of the system is the ontic microstate $\omega \in \Sigma$, while the projective point $\pi(\omega)$ gives the associated epistemic description in the quantum-effective sector. Hamiltonian flow on (Σ, ω_Σ) is therefore interpreted as the underlying physical dynamics, and the projective formalism is understood as its operational image.

For the purposes of this paper, only the following structural facts are required:

- Σ is a finite-dimensional symplectic manifold carrying deterministic Hamiltonian flow.

- $\mathbb{CP}^{n-1}$ is the compact operational pure-state manifold of the corresponding finite-dimensional quantum-effective sector.
- The Fubini–Study measure μ_{FS} is the unique $\mathrm{SU}(n)$-invariant operational measure on $\mathbb{CP}^{n-1}$.
- The projection $\pi : \Sigma \to \mathbb{CP}^{n-1}$ links the ontic and epistemic layers.

These facts are sufficient for the later construction of measurement partitions, geometric outcome weights, and the operational recovery of Born-form probabilities. No ad hoc cutoff is required in this section, and no stronger claim about field theory or many-body limits is needed here.

5.2 Microstate and Evolution

In geometric quantum mechanics, time evolution is represented as Hamiltonian flow on the relevant state manifold. At the operational level, observables correspond to real-valued functions on projective state space, and the Hamiltonian generates a vector field through the symplectic structure. In CSD, this structure is retained but reinterpreted ontically: the underlying deterministic evolution occurs on the symplectic manifold Σ.

Let

$$H_\Sigma : \Sigma \to \mathbb{R}$$

be the ontic Hamiltonian. The associated Hamiltonian vector field X_{H_Σ} is defined by

$$\iota_{X_{H_\Sigma}} \omega_\Sigma = dH_\Sigma,$$

where ω_Σ is the symplectic two-form on Σ. The integral curves of X_{H_Σ} define a smooth flow ϕ_t on Σ satisfying

$$\frac{d\omega}{dt} = X_{H_\Sigma}(\omega), \qquad \omega(t) = \phi_t(\omega_0).$$

Because the symplectic form is closed and non-degenerate, the associated Liouville measure μ_Σ is preserved under the flow. Thus, for any measurable region $\Omega \subset \Sigma$,

$$\mu_\Sigma(\phi_t(\Omega)) = \mu_\Sigma(\Omega).$$

Hamiltonian evolution on Σ is therefore deterministic, invertible, and measure-preserving.

CSD adopts this structure directly and interprets it literally. Each point

$$\omega \in \Sigma$$

is taken to be a physically real microstate, not merely a coordinate representative of a ray. Time evolution is then the actual trajectory of that microstate under the ontic Hamiltonian flow. More precisely:

- ω follows a real trajectory on Σ, not merely a formal one.
- At every instant there is exactly one ontic microstate ω.
- The evolution is deterministic and uniquely fixed by the Hamiltonian.
- The invariant measure μ_Σ is preserved, so the corresponding geometric outcome weights remain stable under isolated evolution.

On this interpretation, the geometric laws used in GQM are not only a reformulation of the operational formalism, but the underlying dynamical laws governing the ontic microstate. The projective and Hilbert-space descriptions are secondary: they encode what can be inferred about the location of ω through the epistemic projection, but they do not replace the ontic trajectory itself.

Within standard GQM, Hamiltonian flow on projective state space is mathematically equivalent to unitary Schrödinger evolution in Hilbert space. In CSD, this equivalence is preserved at the operational level but given a different interpretation. Schrödinger evolution is not taken to be the fundamental law of a physically indeterminate wavefunction. Rather, it is the projective representation of an underlying deterministic Hamiltonian flow of ontic microstates on Σ. The linear unitary dynamics of the wavefunction belong to the epistemic layer; the actual physical motion is the trajectory of ω.

The wavefunction is therefore not the ontic state itself. It is a projective coordinate description of the epistemic image induced by the projection map π. The statistical structure associated with standard quantum theory is not attributed to stochastic ontic evolution, but to invariant geometric weights defined on preparation and outcome regions of Σ.

In summary, Hamiltonian flow on Σ provides the deterministic backbone of the theory. It is single-valued, measure-preserving, and ontic. The operational quantum evolution seen in the standard formalism arises as its epistemic image, while the statistical regularities of quantum mechanics are grounded in the invariant geometry of the underlying flow.

5.3 The Volume Measure μ

In the standard geometric formulation, the pure-state manifold is $\mathbb{CP}^{n-1}$, equipped with the Fubini–Study metric and its associated invariant volume measure. In CSD, $\mathbb{CP}^{n-1}$ is not itself taken to be the ontic state space. Rather, it is the epistemic image of an underlying ontic manifold Σ under a surjective projection

$$\pi : \Sigma \to \mathbb{CP}^{n-1}.$$

The projective manifold carries the transitive isometric action of $\mathrm{SU}(n)$, and this symmetry fixes the unique invariant operational measure on $\mathbb{CP}^{n-1}$.

More precisely, the Fubini–Study metric is induced from the Hilbert-space inner product and defines the standard projective geometry of pure states. Its associated Riemannian volume form determines the Fubini–Study measure μ_{FS}, which is the unique normalised $\mathrm{SU}(n)$-invariant measure on projective state space. In the usual GQM literature, this measure plays a formal role: it is used to define geometric averages and invariant volume, but its physical interpretation is left open.

CSD retains this geometric result but reinterprets it physically. At the ontic level, the manifold Σ carries its own Liouville measure μ_Σ induced by the symplectic form ω_Σ. The relation between the ontic and epistemic layers is required to satisfy a compatibility condition: in the quantum-effective sector, the pushforward of μ_Σ under π reproduces the standard operational measure on projective space, up to the normalisation fixed by finite total volume. Thus for measurable sets

$$C \subset \mathbb{CP}^{n-1},$$

one imposes

$$\pi_*\mu_\Sigma(C) = \mu_\Sigma\big(\pi^{-1}(C)\big) \propto \mu_{\mathrm{FS}}(C).$$

With the normalisation fixed, the induced operational measure is identified with μ_{FS}.

This distinction matters. The invariant measure on projective space is a theorem of the geometric formalism. The ontic interpretation of the corresponding invariant measure structure is an additional claim of CSD. In the present framework, probability is not introduced as a primitive stochastic ingredient. Rather, geometric outcome weights are first defined on Σ by invariant volume ratios of measurable regions. These weights are then mapped to the projective operational description through the projection π. In the quantum-effective sector, Born-form probabilities arise only after the standard admissibility and operational consistency conditions are imposed on the projective probability functional.

Accordingly, the central claims are these:

- the geometry of projective state space fixes a unique invariant operational measure;
- the ontic manifold carries a conserved Liouville measure preserved by Hamiltonian flow;
- the projection map links these two structures in the quantum-effective sector;
- and the resulting invariant geometric weights provide the basis for the operational probability assignments.

The properties of the relevant invariant measures follow from the geometric framework:

- **Symmetry:** μ_{FS} is invariant under the transitive isometric action of $\mathrm{SU}(n)$ on $\mathbb{CP}^{n-1}$.
- **Normalisability:** compactness of the finite-dimensional projective state space ensures finite total operational volume.
- **Dynamical preservation:** the ontic Liouville measure μ_Σ is preserved by Hamiltonian flow on Σ.

CSD does not alter these mathematical facts. Its contribution is interpretive: it treats the invariant measure structure as physically significant rather than merely formal. In that sense, the geometric weights used later in the paper are understood as arising from conserved ontic volume, while the Born-form quantities appearing in the standard formalism are their operational image on projective state space.

The precise geometric conditions under which the pushforward relation

$$\pi_*\mu_\Sigma = \mu_{\mathrm{FS}}$$

holds globally are not derived in the present paper. Here this compatibility is imposed as part of the quantum-effective sector structure. A deeper analysis of the conditions on Σ and π under which this relation holds is deferred to later work.

5.4 From ψ to ω: The Correspondence

The relationship between the epistemic state $|\psi\rangle$ and the ontic microstate ω is mediated by the projection map

$$\pi : \Sigma \to \mathbb{CP}^{n-1},$$

introduced earlier. The epistemic ray $[\psi]$ corresponds to the projected image $\pi(\omega)$, and the conventional wavefunction serves as a coordinate representative of that projective description. The essential point is that $[\psi]$ does not, in general, determine ω uniquely. Rather, it encodes a coarse-grained description of the ontic state. Distinct ontic microstates may project to the same ray, so preparation uncertainty is represented not by ontic superposition but by the many-to-one character of the projection.

Accordingly, the ontic and epistemic layers play different roles. The microstate always evolves on Σ, not on projective state space. What appears in the operational formalism is only the projected description. For a given projective point $[\psi] \in \mathbb{CP}^{n-1}$, the corresponding ontic fibre is

$$\pi^{-1}([\psi]) \subset \Sigma.$$

This fibre may contain many ontic microstates. The wavefunction therefore functions as an epistemic label for a coarse-grained class of ontic states rather than as the ontic state itself.

For a preparation region $\Omega_0 \subset \Sigma$, the associated epistemic description is the image

$$\pi(\Omega_0) \subset \mathbb{CP}^{n-1}.$$

Any representative $|\psi\rangle$ with

$$[\psi] \in \pi(\Omega_0)$$

summarises incomplete information about which ontic microstate in Ω_0 is actually realised. In this sense, the mapping is many-to-one:

$$\pi^{-1}([\psi]) \subset \Omega_0, \qquad |\pi^{-1}([\psi])| \geq 1$$

whenever the preparation procedure does not fix the ontic state completely.

The deterministic dynamics on Σ induce the familiar projective quantum evolution after projection. If the ontic trajectory satisfies

$$\omega(t) = \phi_t(\omega(0)),$$

then the corresponding epistemic trajectory is

$$[\psi(t)] = \pi(\omega(t)).$$

In the quantum-effective sector, the composed map $\pi \circ \phi_t$ reproduces the standard unitary Schrödinger evolution on $\mathbb{CP}^{n-1}$. The wavefunction therefore inherits its dynamics from the underlying Hamiltonian flow, rather than generating that flow at the ontic level. This is the basic sense in which superposition and interference belong to the epistemic representation while the ontic dynamics remain single-valued and deterministic.

Measurement is defined fundamentally on Σ. If a prepared region $\Omega_0 \subset \Sigma$ admits a measurable partition

$$\Omega_0 = \bigsqcup_i \Omega_i,$$

then the corresponding epistemic images are

$$C_i = \pi(\Omega_i) \subset \mathbb{CP}^{n-1}.$$

The ontic geometric weight of outcome i is

$$P(i) = \frac{\mu_\Sigma(\Omega_i)}{\mu_\Sigma(\Omega_0)}.$$

Equivalently, in the quantum-effective sector and after normalisation of the induced operational measure,

$$P(i) = \frac{\mu_{\mathrm{FS}}(C_i)}{\mu_{\mathrm{FS}}(C_0)}.$$

Born-form probabilities arise only after the admissible operational probability functional on projective state space has been fixed. The appearance of the usual Hilbert-space amplitudes therefore reflects the geometry of the epistemic representation, not any ontic superposition of physical states.

In summary, the mapping

$$\omega \mapsto [\psi]$$

provides a principled correspondence between deterministic evolution on Σ and the operational probabilistic structure of quantum mechanics. The wavefunction encodes epistemic restriction, outcome weights are grounded in invariant ontic volumes, and the familiar projective dynamics arise as the image of the underlying Hamiltonian flow.

Note on repeated preparations and typicality: The typicality result used later concerns repeated experiments, each beginning from an initial microstate lying somewhere within a preparation region $\Omega_0 \subset \Sigma$. At any given moment there is only one actual ontic microstate ω, but a preparation procedure does not, in general, determine its exact location within Ω_0. Repeating the preparation therefore yields different ontic initial conditions compatible with the same epistemic description. The phrase "almost all sequences" refers to sequences of such repeated preparations, evaluated with respect to the invariant measure on Ω_0. In this way, the typicality reasoning parallels the familiar logic of classical statistical mechanics: the probabilistic description reflects limited knowledge of the exact ontic microstate, not fundamental stochasticity in the underlying dynamics.

5.5 Geometric Observables

In geometric quantum mechanics, observables are represented by smooth real-valued functions on projective state space. For a Hermitian operator $\hat{A}$, the corresponding geometric observable is

$$f_A([\psi]) = \frac{\langle\psi|\hat{A}|\psi\rangle}{\langle\psi|\psi\rangle}.$$

The symplectic structure determines the associated Hamiltonian vector field X_{f_A} through

$$\iota_{X_{f_A}}\omega_{\mathrm{FS}} = df_A,$$

so that observables generate flows on the operational state manifold. Schrödinger evolution is the special case in which the observable is the Hamiltonian itself. In this way, expectation values, commutator structure, and uncertainty relations all admit geometric expression within the standard GQM framework.

This formalism is mathematically complete, but by itself it remains interpretively neutral. In ordinary GQM, the functions $f_A([\psi])$ are geometric representatives of expectation values on projective state space. The framework does not specify whether they correspond to definite physical properties of individual systems or only to quantities associated with the operational state description.

Constraint-Surface Dynamics preserves this mathematical structure at the operational level, but adds an ontic interpretation beneath it. In CSD, the underlying system occupies a definite ontic microstate

$$\omega \in \Sigma,$$

and projective observables arise as epistemic images of functions defined on the ontic manifold. Thus the operational quantity associated with $\hat{A}$ is not fundamental, but is induced from the deeper geometric structure on Σ through the projection map π.

At the ontic level, one therefore considers real-valued functions

$$F_A : \Sigma \to \mathbb{R},$$

whose projected operational description is represented by the usual projective observable f_A in the quantum-effective sector. The interpretation is that the system possesses a definite ontic microstate throughout, while the standard expectation-value expression belongs to the epistemic description of that state after projection. Operational averages are recovered by integrating the corresponding induced quantity over the relevant preparation region with respect to the invariant measure structure, but the ontic dynamics themselves remain single-valued.

This distinction is important. In GQM, the function on projective state space is part of the operational formalism. In CSD, the corresponding ontic observable is a function on Σ, while the projective expression records only the epistemic image of that deeper structure. The operational expectation values of standard quantum mechanics are therefore not taken to be fundamental properties of an indeterminate state, but summaries of the projected geometric description associated with the underlying ontic dynamics.

Illustrative example: For a two-level system, the operational pure-state space is

$$\mathbb{CP}^1 \cong S^2,$$

the Bloch sphere. In the standard geometric description, a Hamiltonian proportional to σ_z induces the familiar energy function on projective state space, depending on the polar angle θ relative to the z-axis. Operationally, this quantity is interpreted as the expected value associated with the projective state. In CSD, the same projective expression is retained at the operational level, but its physical interpretation is shifted: it is the epistemic image of a deeper ontic observable defined on Σ, evaluated through the projection map. The probabilistic description arises only because the precise ontic microstate is not operationally accessible.

Both GQM and CSD inherit the $\mathrm{SU}(n)$ symmetry of the projective manifold. The group acts transitively and isometrically on $\mathbb{CP}^{n-1}$, so no orthonormal basis is kinematically preferred. In GQM this is a structural fact about the formalism. In CSD, it is preserved as a physical principle at the operational level. Apparent basis selection in an experiment is not attributed to any privileged axis in the state space itself, but to the specific interaction Hamiltonian and the context-defined partition structure induced by the measurement arrangement.

Thus, GQM supplies the geometric representation of observables, while CSD supplies the ontic interpretation beneath that representation. The geometry remains unchanged, but its role is no longer merely formal: observables at the projective level are understood as epistemic images of definite ontic structure on Σ.

6 Dynamics and Physical Isolation

6.1 Hamiltonian Flow on Σ

Geometric quantum mechanics shows that pure-state dynamics can be represented as Hamiltonian flow on the finite-dimensional projective manifold associated with the quantum-effective sector. At the operational level, observables are represented by real-valued functions on projective state space, and the Hamiltonian determines a vector field through the symplectic structure. In CSD, this same dynamical structure is retained, but its underlying ontic realisation is taken to be the symplectic manifold Σ.

Let

$$H_\Sigma : \Sigma \to \mathbb{R}$$

be the ontic Hamiltonian, and let ω_Σ denote the symplectic form on Σ. The corresponding Hamiltonian vector field X_{H_Σ} is defined by

$$\iota_{X_{H_\Sigma}} \omega_\Sigma = dH_\Sigma.$$

Its integral curves determine a continuous flow

$$\frac{d\omega}{dt} = X_{H_\Sigma}(\omega), \qquad \omega(t) = \phi_t(\omega_0),$$

which gives the deterministic evolution of the ontic microstate.

Liouville's theorem implies that this Hamiltonian flow preserves the Liouville measure μ_Σ:

$$\mu_\Sigma(\phi_t(\Omega)) = \mu_\Sigma(\Omega)$$

for every measurable region $\Omega \subset \Sigma$. Hence the total invariant measure is conserved throughout isolated evolution. In the quantum-effective sector, the projected dynamics reproduce the usual unitary Schrödinger evolution on $\mathbb{CP}^{n-1}$. The standard Hilbert-space equation

$$\frac{d\,|\psi\rangle}{dt} = -\frac{i}{\hbar}\hat{H}\,|\psi\rangle$$

is therefore interpreted as the operational image of the underlying ontic Hamiltonian flow.

GQM provides the mathematical content of this equivalence: Hamiltonian flow on the appropriate state manifold corresponds to unitary Schrödinger evolution. What it does not do is assign ontological status to the points moving along that flow. In the standard geometric formulation, those points are elements of the formal state description. In CSD, by contrast, they are taken to represent actual physical microstates.

Accordingly, CSD interprets the Hamiltonian flow on Σ literally:

- ω follows a real trajectory on the ontic manifold, not merely a formal one.
- At every instant there is exactly one ontic microstate ω.
- The evolution is deterministic, governed by the unique vector field X_{H_Σ} fixed by the Hamiltonian.
- The invariant measure μ_Σ is conserved, ensuring stability of the corresponding geometric outcome weights under isolated evolution.

Under this interpretation, the Schrödinger equation is not a fundamental law of an ontically indeterminate wavefunction. It is the projected coordinate description of a deeper deterministic process, namely the continuous Hamiltonian motion of a definite microstate on Σ. What appears as unitary evolution in Hilbert space is the epistemic image of this underlying ontic dynamics.

This reinterpretation turns the formal flow of GQM into physical law. Reality is taken to evolve through a measure-preserving Hamiltonian flow on a finite-dimensional symplectic manifold. Because the invariant measure is conserved, probabilities are not primitive stochastic quantities at the ontic level. Rather, the statistical structure arises from stable geometric weight ratios defined on preparation and outcome regions of Σ.

CSD therefore converts the geometric equivalence between Hamiltonian flow and Schrödinger evolution into an ontological claim: the system evolves deterministically on Σ, and the standard quantum dynamics are the operational representation of that deeper flow. The mathematical structure remains the same as in GQM. What changes is its physical meaning.

6.2 State Preparation

In geometric quantum mechanics, state preparation is represented operationally as the restriction of the accessible state description to a subset of projective state space. A preparation procedure, such as spin alignment, polarisation selection, or controlled initialisation, determines the class of pure states compatible with the imposed experimental constraints. In this sense, the prepared state is represented geometrically by a region of the relevant state manifold.

Within the standard geometric formalism, this is a mathematical characterisation. The preparation procedure identifies a subset of the operational state space consistent with the imposed conditions, but the formalism does not specify whether any deeper ontic element is selected or occupied. It describes the accessible state description, not what physically exists beneath it.

Constraint-Surface Dynamics retains this geometric picture of preparation but adds an ontological interpretation. A preparation region

$$\Omega_0 \subset \Sigma$$

is not merely an abstract subset of the state manifold. It is the physically accessible region of the ontic configuration space within which the actual microstate of the system lies. At the moment of preparation:

- the ontic microstate satisfies
$$\omega \in \Omega_0$$
as a matter of physical fact;
- the exact location of ω within Ω_0 is not generally known to the experimenter;
- the preparation procedure acts by constraining the admissible ontic region, not by creating a fundamentally new ontic state.

Thus, when a laboratory procedure is said to prepare the system "in the state $|\psi\rangle$," the physical content of that claim is that the preparation apparatus has imposed boundary conditions, fields, alignments, or control constraints that restrict the ontic microstate to a region of Σ compatible with the intended operational description. The microstate itself is not created by the preparation act. What changes is the admissible ontic domain within which it must lie.

This distinction between *constraint* and *creation* is central to the ontology of CSD. In the standard formal language, preparation is often described as producing a new state vector $|\psi\rangle$. In CSD, by contrast, preparation is the narrowing of epistemic uncertainty about an already existing ontic microstate. The relevant laboratory procedure does not create a new element of reality. It identifies the subset of Σ consistent with the imposed preparation conditions.

The contrast may be summarised as follows:

- In GQM, preparation defines an operationally admissible region of state space.
- In CSD, preparation constrains the region of Σ in which the actual microstate ω may lie.
- The wavefunction $|\psi\rangle$ encodes the corresponding epistemic description of that constrained ontic region.
- No new ontic state is generated; only the permissible domain is restricted.

Preparation in CSD is therefore a geometric restriction, not a metaphysical reset. The ontic microstate already exists on Σ. Preparation specifies the conditions under which that microstate is localised within an admissible region and then allowed to evolve deterministically under the Hamiltonian flow relevant to the isolated system.

6.3 Coherence as Physical Isolation

Constraint-Surface Dynamics introduces a physical principle that is not part of the purely formal content of geometric quantum mechanics:

Quantum-effective coherence requires physical isolation.

On this view, a system exhibits coherent quantum behaviour precisely when it is dynamically decoupled from the relevant environmental degrees of freedom. In that regime, the system evolves autonomously on its own ontic manifold Σ_{sys}, governed by its effective Hamiltonian H_{sys}. The ontic microstate satisfies

$$\omega \in \Sigma_{\mathrm{sys}},$$

and follows a deterministic Hamiltonian trajectory generated by H_{sys}. While this isolation is maintained, the full structure of the symplectic flow remains available in the operational description, including phase evolution, interference effects, and the standard coherent phenomena associated with the quantum-effective sector.

When the system interacts with an apparatus or environment, this isolation is lost. The relevant ontic description must then be enlarged to include the coupled degrees of freedom. In an idealised setting this may be represented by a combined manifold

$$\Sigma_{\mathrm{tot}} = \Sigma_{\mathrm{sys}} \times \Sigma_{\mathrm{env}},$$

or, more generally, by a larger non-factorised ontic manifold describing the coupled system. The subsystem no longer evolves autonomously. Decoherence is interpreted as this loss of isolation: information that was effectively confined to the subsystem is dispersed into the larger coupled dynamics, and the operational accessibility of interference is correspondingly lost.

Thus isolation is the physical enabling condition for coherent quantum behaviour. Interference, phase-sensitive evolution, and related quantum-effective phenomena are not taken to be mysterious features added to the dynamics, but consequences of the fact that the subsystem can be treated as evolving approximately independently on its own ontic manifold.

Geometric quantum mechanics, in its standard form, establishes the geometric structure of the state space and shows that unitary evolution may be written as Hamiltonian flow on a compact Kähler manifold. What it does not do is specify the physical conditions under which a given subsystem admits an effectively autonomous geometric description. Its concern is structural rather than environmental. It explains how quantum dynamics can be written geometrically, but not when a subsystem should be treated as coherently evolving in physical practice.

CSD supplements this by identifying the missing physical condition: isolation determines when the geometric dynamics of a subsystem correspond to observable quantum coherence. In this way, the abstract symplectic dynamics of GQM are linked to concrete laboratory conditions.

The double-slit experiment provides a simple illustration.

- In GQM, the photon is described operationally by evolution on the appropriate projective state space, and the interference pattern follows from the geometry of that coherent evolution.
- In CSD, the same operational mathematics is retained, but its physical meaning is sharpened. The photon displays interference because, between source and screen, it is sufficiently isolated from measuring interactions. Its ontic microstate evolves under the autonomous dynamics relevant to the isolated subsystem, and the corresponding phase relations remain operationally available.

If a detector is placed at one slit, isolation is destroyed. The photon becomes coupled to detector degrees of freedom, so the relevant ontic description must be enlarged to include the joint system. The disappearance of interference is then not attributed to a primitive collapse or to fundamental randomness, but to the fact that the subsystem no longer admits an effectively autonomous coherent description. Its evolution is now embedded in a larger coupled dynamics, and the operational conditions required for interference are no longer present.

In this sense, CSD interprets decoherence as de-isolation. Isolation supports coherent quantum-effective evolution. De-isolation produces the loss of operational interference and the emergence of stable outcome correlations. The underlying ontic dynamics remain deterministic throughout; what changes is whether the subsystem can still be treated as an approximately independent coherent sector.

Within the present paper, any relation between interference visibility V and the isolation parameter I should be read as a proposed or model-dependent empirical relation, not as a general theorem of the framework. TN0 fixes the canonical definitions of V and I, but a full first-principles derivation of a relation of the form $V \approx 1 - I$, together with its domain of validity and consistency bounds, remains future work. The significance of the proposal is therefore methodological: it identifies a possible experimentally discriminating signature of the CSD picture whose precise form must be established within explicitly stated dynamical regimes.

6.4 Decoherence as Loss of Isolation

In Constraint-Surface Dynamics, decoherence is understood not as a primitive stochastic or fundamentally irreversible process, but as a deterministic loss of subsystem isolation. When a system that previously evolved autonomously begins to couple to external degrees of freedom, the relevant ontic description must be enlarged from the isolated subsystem manifold Σ_{sys} to a larger manifold describing the coupled dynamics. In an idealised representation this may be written as

$$\Sigma_{\mathrm{tot}} = \Sigma_{\mathrm{sys}} \times \Sigma_{\mathrm{env}},$$

though more generally the coupled ontic structure need not remain globally factorised. The essential point is that, once coupling is present, the subsystem no longer evolves as a closed Hamiltonian sector on its own.

The ontic microstate is then described on the enlarged manifold,

$$\omega \in \Sigma_{\mathrm{tot}},$$

and the total Hamiltonian takes the form

$$H_{\mathrm{tot}} = H_{\mathrm{sys}} + H_{\mathrm{env}} + H_{\mathrm{int}}.$$

This total Hamiltonian generates a single joint Hamiltonian flow on the coupled manifold. The subsystem degrees of freedom can no longer, in general, be treated as dynamically autonomous. The corresponding reduced description on Σ_{sys} is not closed under the full coupled evolution.

Physically, this means that coherent quantum-effective behaviour requires subsystem isolation, and that coupling destroys the conditions under which interference remains operationally available. As the system interacts with the environment, correlations accumulate between subsystem and environment degrees of freedom. The phase relations responsible for interference are not destroyed at the level of the total ontic dynamics, but they are dispersed into the larger coupled structure and become inaccessible at the subsystem level.

From the perspective of an observer using only the reduced subsystem description, this appears as suppression of interference and the emergence of effectively classical definiteness. At the level of the full ontic manifold, however, the joint evolution remains deterministic and measure-preserving. Decoherence is therefore interpreted as the projection of a continuous global flow onto a subsystem that has lost its effective isolation.

Geometric quantum mechanics already provides the mathematical language needed to describe decoherence at the operational level. Interference suppression can be represented through reduced-state structure, loss of operational phase accessibility, and the geometry of coupled evolution. What GQM does not do is assign a physical interpretation to why this loss of coherence occurs in concrete terms. CSD supplies that interpretation by identifying loss of isolation as the underlying mechanism.

The contrast may be stated briefly:

- **GQM:** decoherence is described mathematically as loss of operational interference in the reduced formalism.
- **CSD:** decoherence is interpreted physically as the deterministic consequence of coupling, when the ontic microstate no longer evolves as part of an effectively autonomous subsystem.

The progression from coherent to effectively classical behaviour can therefore be described as a continuous geometric sequence:

1. **Isolation (quantum-effective regime):** The subsystem is dynamically autonomous, with
$$\omega \in \Sigma_{\mathrm{sys}},$$
and evolves under its effective Hamiltonian. Coherent phase structure remains operationally accessible.
2. **Coupling initiation:** The interaction term becomes significant, linking subsystem and environment degrees of freedom. The relevant ontic description expands to the coupled manifold.
3. **Correlation growth:** System and environment trajectories become dynamically intertwined. The reduced subsystem description loses autonomy, and interference becomes operationally inaccessible.
4. **Effective decoherence:** The reduced subsystem description appears probabilistic and classical-like, although the underlying global flow remains deterministic and symplectic.
5. **Possible re-isolation:** If coupling weakens sufficiently, the subsystem may again admit an approximately autonomous description, allowing coherent quantum-effective behaviour to re-emerge.

This sequence may be summarised schematically as

$$\text{Isolation} \longrightarrow \text{Coupling} \longrightarrow \text{Effective Decoherence.}$$

In CSD, decoherence is therefore not a stochastic jump and not a fundamental ontic process distinct from the rest of the dynamics. It is the deterministic consequence of embedding a formerly isolated subsystem into a larger coupled Hamiltonian flow.

6.5 Entanglement Structure

For composite systems, the operational state space is enlarged in the standard way. If subsystems A and B have finite dimensions n_A and n_B, then the corresponding pure-state projective manifold is

$$\mathbb{CP}^{n_A n_B - 1},$$

equipped with its usual Kähler structure. At the operational level, this manifold contains both product rays and non-product rays. The latter correspond to entangled pure states in the ordinary quantum-theoretic sense. Geometrically, separable states form a distinguished lower-dimensional subset, while generic points of the full projective manifold are nonseparable.

This structure has been extensively studied in geometric quantum mechanics. In particular:

- **Nonseparability:** The joint projective manifold contains points that cannot be written as simple products of subsystem rays. These are the geometrically entangled states.
- **Geometric characterisation:** Entanglement may be characterised through the position of a point relative to the separable subset, for example by distance-type constructions, orbit structure, or related geometric invariants.
- **Joint geometric structure:** The symplectic and metric tensors on the joint projective manifold encode the full operational correlation structure of the composite system.

Thus, GQM establishes that entanglement is a genuine geometric feature of the composite projective state space. What it does not do is specify what this geometry represents physically. It describes the structure of correlations, but leaves open whether those correlations arise from something ontic or are merely features of the formalism.

Constraint-Surface Dynamics accepts the operational geometry of the composite system, but interprets it through the ontic manifold. The relevant ontic state space for the composite system is denoted Σ_{AB}. In general, CSD does *not* require Σ_{AB} to factor globally as a Cartesian product of subsystem ontic manifolds. Rather, the central claim is that there exists a single actual ontic microstate

$$\omega \in \Sigma_{AB},$$

and that this single ontic state underlies the full joint operational description of the composite system.

At the present stage of the programme, this effective composite projective structure is adopted as part of the finite-dimensional quantum-effective architecture, not yet derived from deeper ontic first principles; what is claimed here is that CSD supplies a workable non-factorising ontic composition together with operational marginal stability and no-signalling under stated assumptions.

The subsystem descriptions are obtained only after projection to the appropriate epistemic level. At the operational level, one may use the standard joint projective state space together with its subsystem reductions. The crucial point is that the apparent subsystem correlations arise because both subsystem descriptions are derived from one and the same ontic microstate on the joint manifold. Entanglement is therefore interpreted as a feature of the unified ontic configuration, not as a mysterious dynamical linkage between independently existing local ontic states.

On this view, the nonseparable structure of the composite projective manifold is the epistemic image of a deeper ontic unity. The joint system is physically represented by one ontic state on Σ_{AB}, and the correlations seen in measurements on subsystems A and B reflect the fact that both operational descriptions originate from that single underlying configuration. No additional signalling mechanism is introduced, and no superluminal influence is required. The correlation structure is geometric because the ontic state space for the composite system is itself unified.

This yields the following contrast:

- **In GQM,** entanglement is a geometric property of nonseparable points in the composite projective manifold.
- **In CSD,** entanglement is interpreted as the operational image of a single real ontic microstate on the joint manifold Σ_{AB}, together with the nonseparable geometric structure of its epistemic projection.

In particular, Einstein-Podolsky-Rosen type correlations are understood as correlations between measurement outcomes whose operational descriptions derive from one and the same ontic configuration. The appearance of nonlocality arises only if one treats the subsystem descriptions as ontically independent. In CSD they are not ontically independent at the joint level. They are partial epistemic descriptions of one composite ontic state.

Accordingly, entanglement is neither an additional force nor an external signalling process. It is the geometric and ontological consequence of treating the composite system as a unified physical object whose operational projective description is nonseparable.

6.6 Classical Apparatus and Boundary Conditions

In Constraint-Surface Dynamics, the measuring apparatus is part of the same deterministic framework as the microscopic system. It is treated as a macroscopic subsystem of the total ontic manifold Σ_{tot}, with internal degrees of freedom that are strongly coupled to environmental modes. Because of this persistent coupling, the apparatus does not typically admit an effectively isolated coherent description. Instead, its operational behaviour is well approximated by stable classical dynamics, with macroscopically distinct configurations rapidly decohered relative to one another.

Accordingly, the apparatus occupies an ontic state

$$\omega_{\text{app}}(t) \in \Sigma_{\text{app}} \subset \Sigma_{\text{tot}},$$

and this state evolves along a single effective trajectory in the relevant macroscopic regime. Coherent superposition between macroscopically distinct apparatus configurations is not operationally sustained, because those configurations are continually stabilised by strong coupling to environmental degrees of freedom. In this sense, the apparatus behaves classically not because it lies outside the theory, but because it is permanently subject to the loss of isolation discussed in the previous subsection.

When the apparatus interacts with the microscopic system during measurement, the relevant coupled evolution is generated by an interaction Hamiltonian

$$H_{\text{int}}$$

on the joint ontic domain describing the coupled system and apparatus. In an idealised representation this may be written as

$$\Sigma_{\text{sys}} \times \Sigma_{\text{app}} \subset \Sigma_{\text{tot}},$$

with the understanding that the full coupled structure need not remain globally factorised. Through this interaction, stable apparatus configurations define the measurement context and thereby induce a partition

$$\{\Omega_i\}$$

of the relevant prepared region for the system. Distinct macroscopic pointer configurations correspond to distinct outcome regions in the coupled ontic description.

The apparatus itself is not described as branching into multiple simultaneously realised macroscopic states. Rather, it remains on a single ontic trajectory within the full coupled dynamics, while the measurement interaction correlates that trajectory with one definite outcome region of the microscopic system. The observer, whether treated as part of the apparatus or as a further correlated subsystem, is similarly described within the same deterministic evolution and therefore records one definite outcome.

The role of the classical apparatus is thus twofold. First, its persistent environmental coupling makes it effectively classical at the operational level. Second, its stable macroscopic configurations provide the physical boundary conditions that define the measurement partition for the microscopic system. In this way, classical registration is not external to the theory. It is the macroscopic limit of the same deterministic geometric dynamics, operating in a regime of permanent de-isolation.

7 Measurement Theory

7.1 What Is a Measurement?

Operationally, a measurement is a controlled interaction between a previously isolated microscopic system and a macroscopic apparatus. The interaction couples selected degrees of freedom of the system to stable macroscopic variables of the device, such as a pointer position, screen location, or readout state, thereby establishing a correlation between microscopic and macroscopic configurations. Environmental coupling then stabilises that correlation by rendering alternative phase relations operationally inaccessible, so that the resulting record persists and can be observed.

In dynamical terms, the relevant interaction is governed by a total Hamiltonian of the form

$$H_{\text{tot}} = H_{\text{sys}} + H_{\text{app}} + H_{\text{int}},$$

together with whatever additional environmental coupling is relevant to the macroscopic stability of the apparatus. During the interaction interval, the joint Hamiltonian flow correlates the ontic microstate of the system with stable macroscopic apparatus configurations. When the transient interaction ends, the correlation remains encoded in the enlarged coupled dynamics as a definite macroscopic record.

From the standpoint of CSD, measurement is therefore a wholly physical process. It is not an exceptional intervention external to the dynamics, but an interaction within the same deterministic geometric framework that governs all other evolution. What makes it a measurement is that the coupling induces a stable outcome partition and correlates that partition with a macroscopic registration state.

Geometrically, CSD expresses this by a measurable partition of the prepared ontic region

$$\Omega_0 \subset \Sigma$$

into disjoint outcome regions

$$\Omega_0 = \bigsqcup_i \Omega_i, \qquad \Omega_i \cap \Omega_j = \varnothing \text{ for } i \neq j.$$

Each region Ω_i is the ontic counterpart of a distinct operational outcome, that is, the geometric analogue of a stable pointer value or other recorded result.

Because the ontic microstate is always definite, there exists one and only one index k such that

$$\omega \in \Omega_k.$$

This is a geometric fact about the actual ontic state, not a stochastic event. Measurement consists in the physical interaction that establishes the partition and the subsequent epistemic identification of which region contains the microstate. No ontic collapse is required, and no branching of reality is introduced. What changes is the observer's information about the already existing ontic configuration.

This is the step at which CSD departs interpretively from standard GQM. GQM supplies the geometric state-space formalism and the Hamiltonian evolution on that space, but does not by itself say what a measurement is physically. CSD adds a concrete ontological account: the system-apparatus interaction induces a partition of the prepared region, and the registered outcome is the revelation of which element of that partition contains the real ontic microstate.

The measurement process can therefore be summarised in three stages:

- **Before measurement:** the ontic microstate lies somewhere in the preparation region Ω_0, but its exact location is not known to the observer.
- **During measurement:** system-apparatus coupling induces a finite measurable partition $\{\Omega_i\}$ of the relevant prepared region.
- **After measurement:** the ontic microstate lies in one definite region Ω_i, and the apparatus enters a stable macroscopic configuration correlated with that region.

Nothing discontinuous occurs at the ontic level. The event is the physical establishment of a partition and the epistemic revelation of an already determined geometric fact about the system.

In this way, CSD adds what the purely formal geometric account leaves open: a physically grounded description of outcome formation. It joins three layers into one framework: the geometric state-space structure inherited from GQM, the deterministic Hamiltonian evolution of the ontic microstate, and the epistemic update associated with outcome registration. GQM explains how the formal state evolves; CSD proposes an account of what a measurement is.

7.2 Decoherence and the Emergence of Measurement Structure

The standard geometric and decoherence formalisms already provide the mathematical tools needed to describe how measurement-like behaviour emerges through environmental coupling. In the usual account, three structural ingredients are central.

1. **Environmental coupling:** The system interacts with external degrees of freedom, so that the relevant description expands from an isolated subsystem to a larger coupled manifold. In an idealised representation one writes

$$\Sigma_{\text{tot}} = \Sigma_{\text{sys}} \times \Sigma_{\text{env}},$$

with total Hamiltonian

$$H_{\mathrm{tot}} = H_{\mathrm{sys}} + H_{\mathrm{env}} + H_{\mathrm{int}}.$$

2. **Suppression of interference:** At the operational level, the subsystem description loses access to phase-coherent cross terms once environmental correlations accumulate. This is the geometric counterpart of the usual reduced-state decoherence story: interference becomes operationally inaccessible in the reduced description.
3. **Emergence of stable outcome structure:** Certain effective configurations remain dynamically robust under environmental coupling. These play the role of pointer states or stable macroscopic alternatives, and the reduced description becomes organised around them.

All of this is well established in the standard geometric and decoherence literature. What those formalisms do not settle is the ontological meaning of the resulting structure. They describe how interference becomes inaccessible, but not what this means physically for the system, the apparatus, or the emergence of definite outcomes.

Constraint-Surface Dynamics accepts the same mathematical pattern but gives it a physical interpretation. In CSD, decoherence is the process by which stable measurement partitions arise from loss of subsystem isolation. More precisely:

- As isolation is lost, the subsystem must be described within the larger coupled manifold Σ_{tot}.
- The ontic flow, which previously evolved as an effectively autonomous subsystem dynamics, becomes embedded in the larger joint dynamics.
- This interaction induces a decomposition of the relevant preparation region into effective outcome regions

  $$\{\Omega_i\},$$

  each associated with a stable macroscopic registration configuration.
- These regions become dynamically robust in the sense relevant to the measurement context: the coupled dynamics suppress operational access to interference between them and stabilise their macroscopic distinction.

Decoherence therefore provides the mechanism of partition formation required by the measurement theory of CSD. In Section 5.1, measurement was defined as the revelation of which outcome region contains the ontic microstate. Here the missing step is supplied: environmental coupling generates the conditions under which such outcome regions become stable and operationally distinct.

On this view, the result of decoherence is not fundamental randomness and not an ontic mixture. It is a structured geometric decomposition of the relevant preparation domain. The original preparation region

$$\Omega_0$$

is transformed, in the measurement context, into a finite family of disjoint outcome regions

$$\Omega_0 = \bigsqcup_i \Omega_i,$$

whose operational distinction is stabilised by the coupled dynamics. The transition from coherent behaviour to effectively classical registration is therefore interpreted as the consequence of de-isolation, exactly in line with the isolation principle introduced earlier.

This yields the central physical principle of CSD measurement theory:

> Quantum-effective coherence requires isolation. Measurement begins when that isolation is broken in such a way that stable outcome partitions are formed.

The contrast with the standard account may be stated succinctly:

- **In standard decoherence theory,** environmental coupling explains why interference becomes inaccessible in the reduced description.
- **In CSD,** the same coupling is interpreted as the physical mechanism by which stable geometric outcome partitions are formed on the ontic manifold relevant to the coupled system.

Thus, decoherence does not itself select a random outcome. Rather, it creates the geometric conditions under which measurement outcomes are well defined: stable, dynamically robust outcome regions correlated with macroscopic apparatus configurations. CSD therefore treats decoherence not merely as a formal suppression process, but as the physical route by which deterministic geometry acquires measurement structure.

7.3 Definite Outcomes Explained

Constraint-Surface Dynamics addresses the measurement problem by adopting a simple ontological principle: the ontic microstate ω always has a definite location on the configuration manifold Σ.

At every instant, the system occupies one and only one ontic point in Σ. When a measurement interaction induces a partition of the relevant preparation region,

$$\Omega_0 = \bigsqcup_i \Omega_i, \qquad \Omega_i \cap \Omega_j = \varnothing \text{ for } i \neq j,$$

the actual microstate lies in exactly one of these outcome regions:

$$\omega \in \Omega_i$$

for one unique index i. This is a geometric fact about the ontic state, not a probabilistic statement. The system is definite throughout. What changes in measurement is the observer's epistemic access to which region contains ω.

On this view, measurement does not create an outcome and does not collapse an indeterminate physical state. It reveals which outcome region already contains the ontic microstate. Definite outcomes are therefore explained by ontic definiteness together with context-defined partition structure.

Because the ontic microstate is definite, the observer's experience of a single outcome follows naturally. The observer, the apparatus, and the system all belong to the same deterministic coupled dynamics. The apparatus becomes stably correlated with one outcome region, and the observer's recorded state becomes correlated with that same region. There are no multiple simultaneously realised outcomes and no ontic branching of reality. Each realised record corresponds to one continuous Hamiltonian evolution on the relevant ontic manifold.

The appearance of randomness arises only from epistemic ignorance prior to outcome registration. Before measurement, the observer knows only that the ontic microstate lies somewhere in the preparation region Ω_0. After the measurement interaction has defined the partition, the observer learns which Ω_i contains the system. The corresponding outcome weights are determined by the invariant geometric ratios

$$\frac{\mu_\Sigma(\Omega_i)}{\mu_\Sigma(\Omega_0)},$$

which, in the quantum-effective sector and after projection to projective state space, yield the standard Born-form probabilities.

In this way, CSD proposes a unified account of definite outcomes:

- the ontic state ω is always definite;
- the outcome regions $\{\Omega_i\}$ are induced by the measurement interaction;
- measurement is revelation of the realised region, not creation of a new ontic state;
- and the observer records a definite outcome because observer, apparatus, and system participate in the same deterministic geometry.

No collapse postulate, no branching ontology, and no primitive stochastic mechanism are introduced. Outcome definiteness is attributed to the geometric fact that the system always occupies one ontic microstate, while statistical regularities are attributed to invariant measure structure on the preparation and outcome regions.

The geometric formalism of GQM already contains Hamiltonian flow, invariant measure, and projective state-space structure. What it does not contain is an ontological assertion that one actual state is physically realised. CSD adds exactly this realist step. By treating ω as real and singular, it turns the geometric formalism from a purely kinematic representation into a candidate account of how definite outcomes arise in physical reality.

7.4 Integrating GQM and Paper B: From Mathematical Measure to Physical Probability

Within the standard geometric formulation, the operational pure-state manifold for an n-level system is

$$\mathbb{CP}^{n-1},$$

and this manifold carries a unique $\mathrm{SU}(n)$-invariant measure, namely the Fubini–Study measure μ_{FS}. This uniqueness follows from the fact that $\mathrm{SU}(n)$ acts transitively and isometrically on projective state space. At the purely mathematical level, μ_{FS} is therefore the natural invariant volume form associated with the geometry of the pure-state manifold.

In the standard GQM literature, this measure is typically used as a formal geometric object. It supports integration over projective state space, geometric definitions of averages, and the invariant description of state-space structure. By itself, however, the formalism does not assign this measure any direct ontological interpretation.

The framework developed in Paper B 2025b sharpens this point. There the invariant projective measure is not treated as merely one convenient option among many. Rather, under the relevant finite-dimensional symmetry assumptions, it is singled out uniquely. The essential ingredients are:

1. **Projective invariance:** Physical pure states are rays rather than vectors, so the probability assignment must respect the projective structure.
2. $\mathrm{SU}(n)$ **covariance:** Operational outcome weights must be invariant under unitary changes of basis.

Under these constraints, the admissible operational measure on projective state space is uniquely fixed to be the Fubini–Study measure. In this sense, the invariant measure is not merely compatible with the geometry. It is required by the symmetry structure of the finite-dimensional operational theory.

Constraint-Surface Dynamics then takes the next step. It does not change the mathematical result established at the projective level. Instead, it provides an ontological interpretation beneath it. The ontic manifold Σ carries a Liouville measure μ_Σ induced by its symplectic structure, and the projection

$$\pi : \Sigma \to \mathbb{CP}^{n-1}$$

is required, in the quantum-effective sector, to be compatible with the standard operational geometry in the sense that the pushforward of μ_Σ reproduces the Fubini–Study measure after normalisation. Thus, for measurable sets

$$C \subset \mathbb{CP}^{n-1},$$

one has

$$\pi_* \mu_\Sigma(C) = \mu_\Sigma\big(\pi^{-1}(C)\big) \propto \mu_{\mathrm{FS}}(C),$$

with the proportionality fixed by the finite total volume of the relevant preparation domain.

This distinction is important. The branching or outcome regions Ω_i are defined ontically as subsets of Σ, not of projective state space. The operational projective quantities appear only after projection. Thus the relevant ontic outcome weight is

$$P(i) = \frac{\mu_\Sigma(\Omega_i)}{\mu_\Sigma(\Omega_0)},$$

where Ω_0 is the prepared ontic region and $\{\Omega_i\}$ is the measurement-induced partition. In the quantum-effective sector, the corresponding operational description is obtained by pushing this measure structure forward through π, yielding the standard projective weighting in terms of μ_{FS}.

The conceptual synthesis is therefore this:

- **GQM** supplies the invariant projective geometry and its unique $\mathrm{SU}(n)$-invariant operational measure.
- **Paper B** shows that this operational measure is uniquely selected by the symmetry structure of the finite-dimensional theory.
- **CSD** interprets the corresponding measure structure physically by linking projective weights to invariant ontic volume ratios on Σ.

Because the ontic Hamiltonian flow preserves μ_Σ, these geometric outcome weights are stable under isolated evolution. Born-form probabilities are therefore not introduced as a primitive stochastic axiom, but arise through a modular chain: invariant ontic geometry determines the weight ratios, projection induces the unique operational measure on projective space, and the admissible probability functional fixes the corresponding Born-form assignment in the quantum-effective sector.

In this way, the traditional Born postulate is replaced, within the scope of the present framework, by a geometrically grounded account in which operational probabilities are the projective image of invariant ontic volume structure.

7.5 Measurement Sequence (Step-by-Step)

The ontological account of measurement in CSD may be summarised as a sequence of physically continuous stages. The purpose of this sequence is to describe what occurs before, during, and after measurement, and to show how observation is interpreted as the revelation of a definite geometric fact about the ontic microstate ω.

1. **Before measurement: epistemic uncertainty:**
   Before any measurement interaction, the system is treated as effectively isolated and evolves autonomously on its subsystem manifold Σ_{sys} under its effective Hamiltonian flow. The preparation procedure defines a region
   $$\Omega_0 \subset \Sigma_{\mathrm{sys}},$$
   within which the actual ontic microstate lies. At this stage, ω has a definite but unknown location. The uncertainty is epistemic rather than ontic. The corresponding wavefunction $|\psi\rangle$ is the projective representation of this limited knowledge: it summarises the operational description associated with the preparation region, weighted by the induced invariant measure structure.
2. **During measurement: decoherence and partition formation:**
   When the system couples to the apparatus and environment, subsystem isolation is lost. The relevant ontic description expands to a coupled manifold, idealised as
   $$\Sigma_{\mathrm{tot}} = \Sigma_{\mathrm{sys}} \times \Sigma_{\mathrm{app}} \times \Sigma_{\mathrm{env}}.$$
   The interaction Hamiltonian induces a deterministic partition of the prepared region into disjoint outcome regions
   $$\Omega_0 = \bigsqcup_i \Omega_i.$$
   This partition is the geometric consequence of the measurement interaction together with environmental decoherence. Interference between the corresponding outcome channels becomes operationally inaccessible, and each Ω_i is stabilised as a distinct outcome region correlated with a macroscopic apparatus configuration.
   The process is continuous and deterministic. No collapse occurs, and the relevant invariant measure is preserved by the underlying Hamiltonian flow. What emerges is a structured outcome geometry in which each Ω_i defines one possible registered result.
3. **After measurement: revelation of ω:**
   Once the interaction has stabilised the outcome structure, the ontic microstate lies in exactly one outcome region:
   $$\omega \in \Omega_i$$
   for one unique index i. The measurement outcome is the revelation of this fact. The apparatus displays the result associated with Ω_i because its macroscopic configuration has become correlated with that region. No duplication of ontology and no branching of reality is introduced. The ontic microstate remains single throughout, and one definite outcome is registered because the coupled system-apparatus-observer dynamics are correlated with one definite outcome region.
4. **Observer correlation:**
   The observer is part of the same deterministic coupled dynamics. The relevant neural, sensory, and recording states are physical degrees of freedom within the enlarged manifold Σ_{tot}. The observer's state becomes correlated with the same outcome region Ω_i as the apparatus. Thus the experience of "I observed outcome i" is not a separate ontological process. It is the continuation of the same deterministic geometric evolution that already correlates system and apparatus with the realised outcome region.
5. **Epistemic update: conditioning on the revealed region:**
   After the outcome is registered, the observer updates the epistemic description. The prior uncertainty over Ω_0 is replaced by the conditioned description associated with the realised region Ω_i. This update is Bayesian in form, but its ontological basis is geometric: the world has not collapsed, and no stochastic jump has occurred. The observer has simply refined the description to the region now known to contain the actual ontic microstate. All subsequent predictions are then made relative to that updated preparation domain.

In this way, the entire measurement process is described as a continuous deterministic sequence: preparation constrains the admissible ontic region, interaction induces a stable partition, one region contains the actual microstate, apparatus and observer become correlated with that region, and the final epistemic update records the revealed geometric fact.

7.6 Observer as Physical System

In standard quantum mechanics, the observer is often introduced abstractly, either as an external agent or through a separate measurement postulate. In Constraint-Surface Dynamics, by contrast, the observer is treated as a physical subsystem governed by the same general deterministic framework as the system under observation, though typically operating in a different dynamical regime.

The observer's apparatus, recording devices, and neural states belong to the macroscopic domain. In practice, these degrees of freedom are already strongly coupled to environmental modes and are therefore effectively decohered. Their behaviour may accordingly be approximated by stable classical variables. In this sense, the observer is not described as an isolated coherent subsystem evolving autonomously on the same effective manifold as the microscopic system. Rather, the observer functions as a dynamically anchored macroscopic subsystem whose stable state helps define the boundary conditions relevant to the measurement interaction.

During measurement, the coupling between system and apparatus destroys the subsystem isolation of the prepared microscopic region $\Omega_0 \subset \Sigma$ and induces the outcome partition

$$\{\Omega_i\}.$$

The ontic microstate of the coupled system lies in exactly one of these regions, and the apparatus enters a macroscopic configuration stably correlated with that region. From the standpoint of the observer, a definite result appears because the classical record of the apparatus corresponds to the unique outcome region containing the ontic microstate. No collapse is required, and the observer is not relocated on the microscopic manifold. What changes is the isolation structure of the coupled system under the influence of the measurement interaction and the already stable macroscopic boundary conditions supplied by the apparatus and observer.

This viewpoint resolves the usual subject-object contrast in three layers:

- **Ontically,** the underlying dynamics on Σ remain continuous and deterministic.
- **Operationally,** classical observers interact with microscopic systems through measurement couplings that alter the partition structure relevant to accessible outcomes.
- **Epistemically,** probabilities arise only because the observer lacks exact knowledge of which outcome region contains the ontic microstate prior to registration.

The observer is therefore neither privileged nor external. The observer is a causally embedded macroscopic subsystem whose effective classical stability makes measurement possible. Observation is the controlled de-isolation of a microscopic subsystem together with the formation of stable correlations between ontic outcome regions and macroscopic records.

In this way, CSD removes the sharp conceptual discontinuity between quantum and classical description. Both are treated as arising from the same geometric foundation. What differs is the dynamical regime: coherent quantum-effective behaviour requires isolation, whereas macroscopic classical stability arises in permanently de-isolated systems whose interference structure is no longer operationally accessible.

7.7 No Preferred Basis

One of the central structural results of geometric quantum mechanics is that no orthonormal basis in Hilbert space is geometrically privileged. The projective manifold

$$\mathbb{CP}^{n-1}$$

is homogeneous under the natural action of $\mathrm{SU}(n)$: any orthonormal basis can be transformed into any other by an isometry of the manifold. The Fubini–Study metric and the associated invariant measure remain unchanged under this action. At the purely geometric level, all bases are therefore equivalent.

This is a mathematical statement in standard GQM. Constraint-Surface Dynamics preserves the same symmetry, but gives it a direct physical interpretation. Because $\mathrm{SU}(n)$ acts transitively on the operational state space and preserves the invariant measure structure, no measurement basis corresponds to a deeper ontic structure than any other. Different measurement bases correspond instead to different context-defined partitions of the same underlying ontic manifold.

In CSD, the outcome regions

$$\{\Omega_i\}$$

are subsets of the ontic manifold Σ, not of $\mathbb{CP}^{n-1}$. This is essential. The ontic microstate ω always evolves on Σ under Hamiltonian flow, and a measurement outcome corresponds to the region of ontic states that lead the apparatus to record a given result. The projective description alone cannot identify the actual ontic microstate. It only constrains ω to lie in the fibre associated with the relevant epistemic state under the projection map. Thus, the probabilities relevant to a given measurement context arise from the invariant geometric weights of the corresponding ontic outcome regions, not from basis structure taken as ontically fundamental.

The physical meaning of basis choice is therefore operational rather than metaphysical. Choosing an observable amounts to choosing the interaction Hamiltonian and hence the partition structure that the measurement apparatus imposes on the prepared ontic region. Different choices of observable correspond to different partitions

$$\{\Omega_i\}$$

of the same ontic geometry. What changes is the measurement context, not the underlying reality.

This yields a principle of basis neutrality:

- all operational observables correspond to admissible partitions of the same underlying ontic manifold;
- the invariant measure structure is the same across all such contexts;
- no basis is ontically preferred, even though different apparatus couplings define different physically realised partitions.

The contrast with standard GQM may be stated simply:

- **In GQM,** basis equivalence is a geometric symmetry of projective state space.
- **In CSD,** the same equivalence is interpreted physically: no measurement direction or observable is built more deeply into reality than any other. What is physically selected in experiment is the context-defined partition, not a preferred basis inscribed in the ontology.

The act of choosing a measurement basis is therefore not the discovery of a hidden preferred structure, but the operational choice of which partition of Σ will define the possible outcomes. Each measurement context induces its own geometric decomposition of the prepared region, while the underlying deterministic flow and invariant measure remain unchanged.

In this way, CSD preserves the basis democracy already present in GQM, but interprets it as a physical symmetry principle. The universality of Born-form probabilities across different measurement contexts is then traced to the invariance of the underlying geometric structure rather than to any privileged orientation in state space.

7.8 Sequential Measurements

In geometric quantum mechanics, successive measurements are represented operationally as successive changes in the projective description of the system, separated by intervals of Hamiltonian evolution on the relevant state manifold. Each measurement context defines its own decomposition of the operational state space, and the formalism correctly reproduces the corresponding transition probabilities and expectation-value updates. What standard GQM does not specify is what physically happens when such measurements occur in sequence.

Constraint-Surface Dynamics embeds sequential measurement into the same deterministic ontological picture used for single measurements. The ontic microstate

$$\omega(t) \in \Sigma$$

always evolves continuously under Hamiltonian flow. A measurement does not interrupt that evolution by collapse. Instead, each measurement interaction induces a new partition of the relevant preparation region on Σ, and the registered outcome is determined by which region contains the ontic microstate at the relevant time.

Successive measurements therefore consist of alternating stages:

1. **Preparation:** The system is prepared in a region
$$\Omega_0 \subset \Sigma.$$
2. **First measurement:** The first apparatus interaction induces a partition
$$\Omega_0 = \bigsqcup_i \Omega_i^{(1)}.$$
The first outcome is the unique region $\Omega_i^{(1)}$ containing the actual ontic microstate.
3. **Intermediate evolution:** After the first registration, the ontic microstate continues to evolve deterministically under the relevant Hamiltonian flow. The post-measurement epistemic description is now conditioned on the realised region.
4. **Second measurement:** A later apparatus interaction induces a new partition of the then-relevant ontic region,
$$\Omega_i^{(1)} = \bigsqcup_j \Omega_{ij}^{(2)},$$
or more generally a fresh partition of the evolved preparation domain appropriate to the new measurement context.

5. **Further repetition:** Additional measurements proceed in the same way: deterministic ontic evolution punctuated by context-defined partitions and epistemic conditioning.

The crucial point is that sequential measurements do not require ontic state reduction. The microstate remains single and continuous throughout. What changes after each outcome is the epistemic description available to the observer and the measurement context that defines the next partition.

Thus, in CSD, sequential measurement is interpreted as:

- continuous ontic Hamiltonian evolution on Σ,
- successive context-defined partitions induced by apparatus couplings,
- and repeated conditioning of the epistemic description on the realised outcome regions.

This preserves the standard operational structure of quantum theory while replacing collapse-style language with a deterministic geometric account. The probabilities observed in sequential experiments arise from invariant geometric weights on the relevant conditioned regions, together with the induced operational probability rule in the quantum-effective sector.

In this way, CSD extends the single-measurement account naturally to multi-step protocols. A sequence of measurements is not a sequence of ontic discontinuities, but a sequence of deterministic geometric evolutions interleaved with context-dependent partitions and epistemic updates.

7.9 The Observer Problem

Problem: Many interpretations place the observer outside the formalism or assign observation a special status, as though measurement required an extra principle beyond the ordinary dynamics of the theory.

CSD solution: In Constraint-Surface Dynamics, observers are physical subsystems embedded within the same deterministic geometry. They are not external agents. Their states are part of the same Hamiltonian evolution as the system under observation. During measurement, the observer and apparatus become correlated with the system through ordinary physical interaction. The resulting macroscopic record corresponds to the same outcome region Ω_i that contains the ontic microstate relevant to the measured system. Observation is therefore correlation, not intervention. Conscious awareness of a single outcome is interpreted as the macroscopic correlate of this stable correlation structure within one continuous deterministic geometry.

7.10 The Preferred-Basis Problem

Problem: Standard formulations often appear to require some extra rule selecting which observable becomes definite in a given measurement. GQM shows mathematically that all bases are geometrically equivalent, but does not by itself explain the physical meaning of that fact.

CSD solution: Constraint-Surface Dynamics treats basis democracy as a physical symmetry principle. Because $\mathrm{SU}(n)$ acts transitively and preserves the invariant operational measure, no basis is ontologically privileged. What changes from one experiment to another is not the underlying geometry, but the interaction context. Each measurement arrangement induces its own partition $\{\Omega_i\}$ of the relevant ontic region, and that partition defines the operational outcomes. Thus no basis is preferred in nature itself. Experimental context determines which partition is implemented, while the manifold and its invariant structure remain symmetric under all admissible choices.

7.11 The Probability Stability and Frequency Problem

Problem: In a deterministic theory, one must explain how stable statistical frequencies arise and why those frequencies remain time-independent across repeated experiments.

CSD solution: The required stability follows from geometric measure preservation. Hamiltonian flow on Σ preserves the invariant Liouville measure μ_Σ. Hence relative volume ratios of the form

$$\frac{\mu_\Sigma(\Omega_i)}{\mu_\Sigma(\Omega_0)}$$

are stable under the relevant deterministic evolution. These ratios provide the geometric outcome weights, and through the typicality framework they determine long-run frequencies across repeated preparations. Probability conservation is therefore not an additional axiom. It is a direct consequence of volume preservation under Hamiltonian flow.

7.12 The Nonlocality and Bell-Correlation Problem

Problem: Bell-type and EPR-type experiments reveal correlations between distant systems that appear nonlocal when described in ordinary spacetime. Standard realist approaches often seem forced either to introduce superluminal influences or to abandon determinism. GQM describes entanglement geometrically on the composite projective manifold, but does not specify what that geometry represents physically or how such correlations appear as spacetime events.

CSD solution: Constraint-Surface Dynamics addresses this by distinguishing between locality on the ontic manifold and apparent nonlocality in emergent spacetime. For a composite system, the total ontic state is a single definite microstate

$$\omega \in \Sigma_{AB}.$$

This one ontic configuration determines the correlated operational behaviour of both subsystems simultaneously. When projected into the observer's effective spacetime description, the corresponding subsystem descriptions appear spatially separated. However, they are not ontically independent. They are partial epistemic descriptions of one connected configuration on the joint manifold.

The Hamiltonian flow on Σ_{AB} is taken to be locally continuous and measure-preserving. No superluminal signal is introduced at the ontic level. The appearance of spacelike coordination arises because spacetime separation belongs to the emergent projected description, whereas the underlying ontic state is unified. Bell-type correlations are therefore attributed to the geometry of the joint ontic manifold together with context-defined measurement partitions, not to signalling, collapse, or pre-existing local values assigned independently to each subsystem.

In the two-qubit case, the relevant joint ontic region supports a context-dependent partition structure associated with the chosen measurement settings. The corresponding invariant volume ratios are intended, within the companion finite-dimensional analysis, to recover the observed quantum correlation structure at the operational level while remaining compatible with no-signalling. On this account, empirical nonlocality is an emergent appearance in spacetime, whereas the deeper ontic dynamics remain deterministic and geometrically unified.

7.13 The Classical-Apparatus Boundary Problem

Problem: A satisfactory interpretation must explain how a classical measuring apparatus can arise from quantum constituents without invoking an ad hoc cut-off or a fundamentally separate classical domain.

CSD solution: In Constraint-Surface Dynamics, macroscopic apparatus are treated as systems in a regime of persistent de-isolation. Because they are strongly and continually coupled to environmental degrees of freedom, they do not admit an effectively autonomous coherent description in the same sense as isolated microscopic systems. Their relevant macroscopic configurations are therefore stable, effectively classical, and well approximated by single deterministic trajectories in the coupled dynamics. The apparatus enters the measurement process not as a superposed quantum object requiring collapse, but as a stable macroscopic subsystem that defines the interaction context and thereby the partition structure imposed on the microscopic preparation region. In this way, the apparatus provides the physical boundary conditions for measurement without requiring a separate ontological category or an ad hoc quantum-classical cut-off.

8 Comparison to Alternative Approaches

The following sections compare the present framework with the principal interpretive families of quantum theory, highlighting both points of alignment and points of distinction.

CSD is ψ-epistemic in the restricted sense that the operational ray $[\psi]$ is not the ontic microstate itself but the image of a deeper ontic state under a many-to-one projection $\pi : \Sigma \to \mathbb{CP}^{n-1}$. In that respect it differs from ψ-ontic reconstructions targeted by PBR-type arguments, because the CSD compositional ontology does not assume the Cartesian product-lift structure used in preparation-independence theorems. It also differs from generalized noncontextual models because outcomes are explicitly context-indexed through measurement interactions, so CSD does not posit a single global value assignment for incompatible observables. Relative to Bohmian mechanics, the framework replaces configuration-space guidance by Hamiltonian flow on an ontic symplectic manifold together with epistemic projection. Relative to Everettian and decoherence-only accounts, it retains deterministic dynamics but attributes each run to a single realised outcome basin rather than to branching multiplicity alone.

Framework	Core Idea	Treatment of Measurement & Outcomes	CSD Comparison	Distinctive CSD Advantage
Copenhagen	Wavefunction ψ is an epistemic tool; collapse produces outcomes probabilistically.	Indeterministic collapse; classical–quantum divide remains undefined.	Reproduces Copenhagen statistics but dispenses with collapse. Outcome = which region Ω_i contains the real microstate ω.	Single deterministic surface Σ eliminates dualism and the collapse postulate.
Many-Worlds (Everettian)	Universal ψ evolves unitarily; each outcome is a real branch.	Branching is permanent; all outcomes exist in parallel.	Keeps determinism but not multiverse ontology. Branching = temporary partitioning of Σ into Ω_i.	Unified Σ allows re-isolation and re-interference, with no irreversible branching.
Bohmian Mechanics	Particles have definite positions guided by ψ.	Dual ontology (ψ + positions); nonlocal guidance equation.	ω on Σ unifies ψ and positions; flow is locally causal on Σ, appearing nonlocal only in spacetime.	Determinism without a separate pilot wave or dual-state ontology.
GRW / CSL Collapse	Random spontaneous collapses ensure macroscopic definiteness.	Adds stochastic terms and violates exact unitarity.	Preserves Hamiltonian flow; outcomes arise from geometric partitioning and decoherence.	No new collapse parameters; effective collapse is replaced by deterministic geometric partitioning of Ω_i.
QBism (Quantum Bayesianism)	ψ represents an agent's beliefs; probabilities are subjective.	Measurement updates personal credence; no objective reality assumed.	CSD is objective and realist: Σ is ontic. ψ encodes ignorance about ω.	Explains why Bayesian updating appears naturally while retaining realism.
Relational QM	Properties exist only relative to other systems.	All quantities are relational; no absolute state.	CSD agrees on relational descriptions but adds a single real surface Σ where relations are geometrically anchored.	Bridges relational epistemology with deterministic ontology.
Decoherence Program	Environment entanglement suppresses interference, producing apparent classicality.	Explains loss of coherence but not selection of one outcome.	CSD treats decoherence as physical loss of isolation: $\Omega_0 \to \{\Omega_i\}$ with Born weights from invariant volumes.	Turns decoherence from approximation into a deterministic geometric mechanism.
Standard QM (Postulates)	Hilbert-space axioms and Born rule taken as given.	Probabilities introduced axiomatically; interpretation optional.	CSD retains empirical equivalence but grounds Born-form weights in symmetry and volume ratios.	Deterministic geometric underpinning replacing primitive probabilistic postulates.

Table 1: Comparison of Constraint-Surface Dynamics with major interpretive approaches to quantum theory:

8.1 Copenhagen

The Copenhagen view treats the wavefunction as a tool for computing probabilities rather than as a physically real entity. Measurement is taken to involve an indeterministic collapse of the wavefunction, producing definite outcomes according to the Born rule.

CSD comparison: Constraint-Surface Dynamics preserves the empirical predictions associated with the Copenhagen framework but replaces its dualism with a continuous deterministic geometry. Collapse is not required. The ontic microstate ω is always definite, and measurement reveals which outcome region Ω_i contains it. Born-form probabilities arise from invariant geometric weight ratios rather than from a primitive collapse rule. In this way, CSD reproduces the operational statistics of the Copenhagen view while dispensing with both the undefined collapse process and the unresolved split between epistemic formalism and physical ontology.

8.2 Many-Worlds

The Everettian 1957 approach treats the wavefunction as describing a universal superposition that never collapses. Each possible outcome corresponds to a branch of reality, and observers experience one branch because their own states become correlated with that outcome. The branch structure is taken to be interpretively fundamental, even though the underlying unitary dynamics remain continuous.

CSD comparison: Constraint-Surface Dynamics retains deterministic evolution but not the ontology of parallel worlds. The manifold Σ remains single and continuous. What appears operationally as branching is interpreted instead as temporary dynamical partitioning of the relevant ontic region into disjoint outcome regions Ω_i. Observers confined to one such region record one definite result, yet the underlying manifold remains unified. Where Many-Worlds multiplies realised worlds, CSD keeps one deterministic geometric surface whose partition structure explains the same measurement appearances.

8.3 Bohmian Mechanics

Bohmian mechanics 1952a restores determinism by supplementing the wavefunction with definite particle positions guided by a pilot wave. It therefore introduces a dual ontology: both ψ and the particle configuration are taken to be physically real, and the guiding equation leads to an explicitly nonlocal spacetime description.

CSD comparison: CSD aims to retain determinism without a dual ontology. The ontic microstate ω on Σ plays the unifying role that Bohmian mechanics 1952a divides between particle configuration and guiding wave. Correlations that appear nonlocal in emergent spacetime are attributed to the geometry of the joint ontic manifold rather than to a separate pilot-wave influence. In this sense, CSD preserves Bohm's deterministic ambition while simplifying the ontology to one continuous geometric structure.

8.4 GRW (Spontaneous Collapse)

The Ghirardi-Rimini-Weber 1986 approach modifies Schrödinger evolution by introducing spontaneous stochastic collapses, thereby ensuring macroscopic definiteness at the cost of exact unitarity and additional parameters.

CSD comparison: Constraint-Surface Dynamics preserves Hamiltonian evolution and does not add stochastic collapse laws. Outcome definiteness arises from geometric partition structure and deterministic de-isolation rather than from random jumps. Apparent collapse is reinterpreted as the revelation of the definite outcome region containing ω. Where GRW changes the dynamics to secure classicality, CSD explains effective classicality as the structural consequence of deterministic geometry plus environmental coupling.

8.5 QBism

Quantum Bayesianism interprets the wavefunction as an agent's personal bookkeeping device for expectations about future experiences. Probabilities are subjective degrees of belief, and quantum theory is taken to be a normative framework for rational updating rather than a description of objective physical reality.

CSD comparison: CSD departs sharply from QBism at the ontological level. It treats Σ as physically real and the ontic microstate ω as objective. Born-form probabilities are grounded in invariant geometry rather than in subjective coherence constraints. However, CSD does retain an epistemic component: the wavefunction ψ encodes incomplete knowledge of the underlying ontic microstate, and measurement leads to a Bayesian-style update of that knowledge once an outcome is registered. In this way, CSD preserves realism while explaining why Bayesian updating appears naturally in the operational description.

8.6 Relational Quantum Mechanics

Relational Quantum Mechanics proposes that physical properties are meaningful only relative to another system. On this view, interactions define relational facts, and there is no single absolute state description valid from all perspectives.

CSD comparison: CSD agrees that measurement outcomes are relational at the descriptive level: each outcome partition $\{\Omega_i\}$ is defined by a specific system-apparatus interaction. However, it adds a deeper ontological claim. All such relations are grounded in one real geometric manifold Σ, and the ontic microstate ω remains single and definite throughout. Thus, properties may be relational in operational description while remaining anchored in a determinate underlying geometry. CSD therefore incorporates the descriptive insight of the relational view while supplying a definite ontic substrate.

8.7 The Decoherence Program

The decoherence programme explains the appearance of classicality through environmental entanglement, showing how interference terms become inaccessible in the reduced description of a subsystem. It provides a powerful effective account of coherence loss, but by itself it does not explain why one definite outcome is observed or why the associated probabilities should be interpreted physically.

CSD comparison: Constraint-Surface Dynamics incorporates the standard mathematical apparatus of decoherence, but embeds it in a deterministic ontology. Decoherence is interpreted as the physical loss of subsystem isolation. As coupling to apparatus and environment grows, the prepared region Ω_0 is partitioned into stable outcome regions $\{\Omega_i\}$, and the corresponding invariant geometric weights supply the Born-form probabilities in the quantum-effective sector. The observer's awareness of a single outcome follows from stable correlation with one such region. In this way, CSD turns decoherence from an effective suppression formalism into a geometric mechanism for outcome formation.

9 Objections and Responses

The following subsections anticipate common objections and clarify where the contribution of CSD lies. Each response situates the framework relative to existing theory, acknowledging mathematical continuity while defending the conceptual significance of the ontological extension.

9.1 "This Is Just Relabelling GQM"

Objection: Geometric Quantum Mechanics already provides a complete geometric reformulation of quantum mechanics. It identifies the projective manifold as the state space, represents observables as real-valued functions, and describes evolution by Hamiltonian flow. On this view, CSD appears merely to rename mathematical points as "real" microstates and to reinterpret existing equations ontologically. If the mathematics is unchanged, why not simply remain within GQM?

Response: It is correct that the mathematical infrastructure of CSD is inherited from GQM, and that debt should be stated plainly. However, the interpretive shift changes the scientific role of the framework. CSD does not propose a new operational formalism. It proposes a new account of what the existing formalism is about. That change is not cosmetic, for four reasons.

1. **It addresses the measurement problem:** GQM describes the geometry of state evolution, but does not explain why a measurement yields one definite outcome. CSD adds an ontic layer in which the microstate ω is always definite, so that measurement becomes revelation of the realised outcome region rather than collapse of an indeterminate state.
2. **It clarifies the status of the wavefunction:** In standard GQM, ψ is part of the formal coordinate description of projective state space, but its ontological status is left open. In CSD, ψ is treated explicitly as epistemic: it encodes the projected description of incomplete knowledge about the underlying ontic microstate.
3. **It gives physical meaning to invariant measure structure:** GQM uses the Fubini–Study measure as the natural invariant operational measure on projective space. CSD, together with the symmetry argument developed in Paper B 2025b, interprets the corresponding invariant measure structure as physically relevant to outcome weights rather than as a merely formal integration tool.
4. **It supplies a realist ontology:** GQM is intentionally neutral about what exists physically. CSD is not. It treats the ontic manifold Σ as physically real and the Hamiltonian flow on that manifold as the actual evolution of the system.

So the issue is not whether the mathematics changes. It does not, at the operational level. The issue is whether adding an ontological interpretation to that mathematics matters. CSD answers yes: the added realist layer turns a geometric reformulation into a candidate physical account of measurement, probability, and definite outcomes. In that sense, the contribution is not a replacement of GQM, but an ontological development of it.

9.2 "This Is Just Bohmian Mechanics in GQM Language"

Objection: Constraint-Surface Dynamics appears to reproduce Bohmian mechanics 1952a in geometric form. It is deterministic, realist, and based on continuous flow on a state manifold. Perhaps it is simply pilot-wave theory rewritten in projective language.

Response: CSD and Bohmian mechanics do share determinism and realism, but they differ in ontology, symmetry structure, and the status assigned to the wavefunction.

1. **No pilot wave or empty branches:** In Bohmian mechanics 1952a, the wavefunction is ontic and guides the particle configuration, including in regions where no particle is actually located. In CSD, by contrast, the wavefunction is not fundamental. The only ontic state is the microstate $\omega \in \Sigma$. The projective wavefunction

description is epistemic, arising from coarse-graining through the projection map. There are therefore no empty waves and no separate guiding field.

2. **No preferred configuration basis:** Bohmian mechanics 1952a privileges a configuration-space ontology based on particle positions and often faces the problem of how to reconcile that structure with broader symmetry demands. CSD is built from the geometric symmetry of the underlying state manifold and retains the basis neutrality of the standard projective formalism. Different observables correspond to different context-defined partitions of the same ontic geometry, not to fundamentally preferred ontic coordinates.
3. **Born-form weights are not introduced through a separate equilibrium postulate:** Bohmian mechanics 1952a typically requires a quantum-equilibrium hypothesis to recover the standard $|\psi|^2$ statistics. CSD instead seeks to ground the operational weighting in invariant geometric measure structure together with the admissibility constraints on the projective probability functional. In that sense, the Born-form rule is not added as an independent equilibrium condition.
4. **Continuity with geometric quantum mechanics:** CSD remains inside the geometric framework already present in GQM. The ontic microstate ω lives on Σ, and the operational quantum formalism arises from the projection to projective state space. The theory is therefore presented not as an additional hidden-variable overlay on top of quantum mechanics, but as an ontological interpretation of existing geometric structure.

For these reasons, CSD should not be identified with Bohmian mechanics 1952a in different notation. Both are deterministic and realist, but CSD replaces the dual ontology of configuration plus pilot wave with a single ontic manifold together with an epistemic projective image. The result is a geometric determinism that is closer in structure to GQM than to pilot-wave theory, even where the two share some philosophical motivations.

9.3 "This Adds No New Predictions"

Objection: If CSD reproduces the empirical predictions of standard quantum mechanics within its intended domain, then it may be interpretively interesting but physically redundant.

Response: The first aim of CSD is conceptual completion rather than immediate empirical novelty. The framework is intended to provide a coherent account of definite outcomes, probability, and measurement without altering the successful operational structure of finite-dimensional quantum mechanics. In that sense, it stands in the same tradition as other interpretation-sensitive developments: explanatory improvement is itself a legitimate theoretical achievement when it resolves structural problems left open by the standard formalism.

That said, CSD is not intended to be empirically vacuous. Its ontological commitments constrain how the framework could fail. In particular, the theory ties operational probability assignments to invariant geometric structure on a finite ontic manifold, links coherence explicitly to isolation, and treats measurement as context-defined partitioning rather than fundamental collapse. If those commitments are extended beyond the present finite-dimensional scope, they point toward possible empirical pressure points involving the limits of decoherence, extreme isolation regimes, or failures of the assumed measure-compatibility structure. The present paper does not claim to derive a novel quantitative deviation within the established finite-dimensional sector. Its claim is more modest: the framework is structured in a way that could in principle be wrong, and later extensions may expose that more sharply.

So the absence of immediate new predictions in the present scope does not render the framework redundant. CSD proposes a different account of what the standard formalism means, and that account has consequences for how later extensions, limits, and possible empirical deviations must be formulated. At minimum, it aims to replace an incomplete axiomatic picture with a deterministic geometric ontology whose explanatory commitments are precise enough to be challenged.

9.4 "This Is Still Nonlocal"

Objection: CSD admits nonlocal correlations in observed spacetime. Is this not merely a reformulation of the usual nonlocality problem?

Response: CSD does not deny the empirical appearance of nonlocal correlation. What it changes is the level at which locality is evaluated. The underlying dynamics are continuous Hamiltonian flow on the ontic manifold Σ, and no superluminal influence or preferred spacetime foliation is introduced there. The apparent nonlocality arises only in the projected spacetime description, where observers treat subsystem descriptions as if they were ontically independent when in fact they are partial views of one unified ontic configuration. In that sense, CSD makes the source of the observed nonlocality explicit rather than mysterious. Local continuity is preserved at the geometric level, while spacetime nonlocality is interpreted as an emergent feature of projection rather than a fundamental causal violation.

9.5 "It Is Just an Interpretation, Not a Theory"

Objection: CSD changes ontology but adds no new operational mathematics or test equations. Why should this count as a theory rather than merely an interpretation?

Response: Interpretive change matters when it resolves explanatory gaps left open by the formalism. CSD does not claim to replace the successful operational mathematics of finite-dimensional quantum mechanics. Its claim is that the existing geometry can be given a definite physical interpretation that closes several structural gaps concerning measurement, outcome definiteness, and probability. In doing so, it changes the explanatory architecture of the theory. The manifold is no longer merely formal, the invariant measure is no longer merely a mathematical integration tool, and measurement is no longer left as an undefined transition from formal evolution to recorded fact. In that sense, CSD is interpretive in method but theoretical in ambition: it seeks to turn an abstract geometric formalism into a physically grounded account of what exists and how observed outcomes arise.

9.6 "You Have Hidden Randomness in Ignorance"

Objection: By treating probabilities as epistemic, CSD may simply be hiding indeterminism behind ignorance.

Response: The ignorance in CSD is not arbitrary and does not function as a vague substitute for unexplained randomness. The probability structure is tightly constrained by invariant geometry. Outcome weights arise from measure-preserving Hamiltonian flow and the relative volumes of preparation and outcome regions. These weights are then linked to operational probabilities through the symmetry-fixed projective measure and the admissible probability functional. Thus the epistemic element is structured by exact geometric constraints. Randomness is not fundamental at the ontic level. What appears probabilistic reflects limited knowledge of the precise ontic microstate together with the invariant geometric structure of the accessible state space.

9.7 "It Is Not Compatible with Relativity"

Objection: If CSD posits a deeper configuration manifold Σ, how can it remain consistent with Lorentz symmetry and relativistic causal structure?

Response: The present formulation is intended to remain compatible with known relativistic constraints at the operational level, not to provide a completed relativistic theory. All measurable predictions in the quantum-effective sector are required to respect the standard relativistic no-signalling structure, and the framework does not introduce observable violations of Lorentz invariance in that domain. What CSD adds is a deeper geometric layer beneath the emergent spacetime description. The claim is therefore one of compatibility rather than full derivation. A full account of how relativistic structure emerges from Σ is deferred to later work. The current framework should be read as preserving empirical Lorentz compatibility while leaving the deeper derivation of spacetime geometry for the next stage of the programme.

9.8 "It Is Philosophical, Not Scientific"

Objection: Since CSD reproduces existing predictions within its present scope, it belongs to philosophy rather than physics.

Response: CSD is physical in at least two clear senses.

1. It supplies an explicit ontology for structures already used in physics, namely the ontic manifold Σ, the invariant measure structure, and the Hamiltonian flow.
2. It yields a definite class of physical expectations about how isolation, decoherence, and partition stability should behave in limiting regimes. Any deterministic geometric completion of quantum statistics is therefore a legitimate physical hypothesis, even when its first contribution is explanatory rather than immediately predictive.

Interpretive clarity is not external to physics. It helps determine what the formalism is about, what counts as a physical process, and where genuine empirical pressure points may arise in later extensions. In that sense, CSD is not offered as philosophy in place of physics, but as an attempt to provide the physical basis on which further empirical development could proceed.

9.9 Compatibility with Additional No-Go Theorems

Several further no-go results are often taken to constrain deterministic or realist completions of quantum mechanics. Within CSD, these theorems do not introduce a new obstruction beyond the structural points already made for Bell, Kochen–Specker, Fine, and PBR. The common reason is that CSD is deterministic at the ontic level but contextual at the level of measurement implementation: outcomes are definite for the realised measurement context, while no single global value assignment is defined across incompatible contexts.

Leggett–Garg: Leggett–Garg inequalities test the conjunction of macroscopic realism and non-invasive measurability. CSD affirms definiteness of the ontic microstate ω, but it does not assume non-invasive measurability. Measurement is a physical interaction that alters the effective partition structure associated with the system–apparatus coupling. A Leggett–Garg violation therefore does not count against CSD. The theorem excludes theories that combine definite states with genuinely non-invasive readout, whereas CSD retains the former and rejects the latter.

Conway–Kochen Free Will theorem: The Conway–Kochen result does not force indeterminism in CSD. The relevant obstruction applies to theories in which outcomes for all admissible settings are fixed by a single globally defined response function of prior information. CSD does not posit such a function. As in the Fine–Bell analysis, the realised context determines which outcome map is physically instantiated. Determinism therefore holds context by context, but not as a global assignment over incompatible measurement choices.

Colbeck–Renner: The Colbeck–Renner theorem is compatible with the intended scope of CSD. That theorem constrains extensions of quantum theory that would yield improved predictive power while preserving the relevant free-choice and no-signalling assumptions. CSD is not proposed as a super-predictive extension of finite-dimensional quantum mechanics. Its claim is instead ontological and geometric: the operational probabilities remain those of standard quantum theory in the quantum-effective sector, while the underlying account of states, measurement, and probability is altered. In that sense, Colbeck–Renner limits empirical enhancement, not ontological completion.

Hardy-type arguments: Hardy-type paradoxes are handled by the same contextuality structure already used for Bell. The contradiction depends on counterfactual combination of outcomes across incompatible settings. In CSD, distinct measurement choices define distinct context-indexed partitions, and there is no requirement that these partitions admit a common global refinement. The paradox therefore does not arise as an internal inconsistency of the theory.

Frauchiger–Renner: Frauchiger–Renner-type contradictions likewise do not go through unchanged in CSD. Every observer is treated as a physical subsystem, and every measurement corresponds to a real partition event on the underlying ontic manifold. A later observer may use an epistemic description that is superpositional relative to incomplete information, but this does not imply that no definite outcome occurred at the earlier stage. The contradiction is avoided because CSD does not identify epistemic superposition with ontic indefiniteness.

No-cloning and no-broadcasting: Standard operational impossibility results such as no-cloning and no-broadcasting are not in tension with CSD. Since the finite-dimensional operational sector reproduces the standard projective and mixed-state probability structure, the usual impossibility of perfectly cloning arbitrary unknown states, and of broadcasting generic noncommuting states, is inherited rather than evaded.

Leggett-type nonlocal hidden-variable inequalities: Leggett-type inequalities constrain a restricted class of nonlocal hidden-variable models. CSD need not belong to that class, because it is not formulated as a theory of globally assigned local properties supplemented by a special nonlocal correction. Its correlation structure is instead carried by the geometry of the joint ontic manifold together with context-dependent partitioning.

Taken together, these results reinforce one structural point. CSD is not a theory of globally pre-assigned values for all conceivable measurements. It is a deterministic, context-dependent geometric completion in which definite outcomes occur for the measurement actually implemented, while incompatible contexts are represented by distinct partition structures rather than by a single universal valuation map.

10 Extensions and Future Directions

The following subsections outline the next stages of the programme, moving from near-term applications to broader conceptual extensions.

10.1 Immediate Applications

Geometric quantum mechanics has already found applications in quantum information, geometric optimisation, and entanglement geometry. Constraint-Surface Dynamics builds directly on this mathematical background, but adds an ontological interpretation that may also be useful in settings where the geometric formalism has so far remained purely descriptive.

1. **Geometric optimisation:** Volume-based reasoning on Σ may suggest new optimisation methods for quantum control, gate compilation, and learning algorithms operating on curved state manifolds. Treating ω as a definite trajectory rather than as an indeterminate amplitude suggests search strategies based on deterministic flow, geodesic structure, and invariant regions of the configuration manifold.
2. **Entanglement geometry:** The realist interpretation of the joint manifold for composite systems may allow entanglement structure to be reformulated in explicitly geometric terms. In particular, it may become possible to define correlation measures using invariant volumes, partition structure, or distance from dynamically admissible separable sectors, rather than relying only on amplitude-based descriptions.
3. **Quantum chemistry and molecular dynamics:** Geometric state-space methods already assist in visualising correlated quantum behaviour. A CSD-style treatment could, in principle, support deterministic descriptions of transition structure on finite configuration manifolds, offering an alternative interpretive picture of molecular evolution without appeal to collapse-based language.

There are several reasons why CSD may add value beyond standard GQM in these settings:

- A realist ontology can guide algorithmic interpretation by treating a computational process as deterministic flow on Σ, which may simplify questions of stability and robustness.
- Volume-based reasoning offers heuristics distinct from amplitude-based interference, potentially allowing conserved geometric quantities to be used as optimisation invariants.
- The isolation principle suggests physically motivated strategies for coherence management, by treating preservation of quantum-effective behaviour as preservation of local dynamical autonomy on Σ.

In this sense, CSD may convert parts of GQM's abstract geometry into a more operational toolkit, in which ontological interpretation helps motivate concrete mathematical and computational developments.

10.2 Relationship to the Geometric Quantum Mechanics Community

The geometric quantum mechanics community is the natural mathematical environment in which CSD should be tested, refined, and criticised. Researchers working in differential geometry, symplectic topology, quantum information, and quantum foundations already possess many of the tools required to sharpen the framework. CSD does not replace that work. It attempts to extend it by adding a physical interpretation to the existing geometric structure.

Three points are particularly relevant:

- **Synergy:** GQM provides the mathematical rigour and formal precision. CSD proposes an ontological reading of that structure and asks what new explanatory or practical value may follow.
- **Collaborative focus:** Productive joint work could include measure theory on projected manifolds, curvature and stability questions, composite-system geometry, and volume-based quantum information quantities.
- **Bridging domains:** CSD sits at the boundary between pure geometric formalism and physical interpretation. It may therefore provide a bridge between abstract work on projective and symplectic manifolds, foundational questions about measurement and probability, and possible future applications in quantum technologies.

More broadly, the future of the programme can be viewed in three layers:

- **Near term:** clarification of the finite-dimensional framework, publication of the supporting notes, and formalisation of the core Born-rule structure.
- **Medium term:** extension toward substrate and effective-theory questions, including the structure of Σ, locality, and large-system limits.
- **Long term:** investigation of whether the same geometric ontology can support broader connections to field theory, spacetime emergence, and gravity.

The guiding idea remains unchanged throughout: quantum theory is interpreted as the operational image of a deeper deterministic, measure-preserving geometric dynamics on Σ. The present paper develops that claim only in the finite-dimensional setting, but it also defines the conceptual route by which the programme might be extended.

11 Conclusion

11.1 What Has Been Achieved

Constraint-Surface Dynamics has been developed as an ontological extension of geometric quantum mechanics within the finite-dimensional quantum-effective sector. GQM supplies the mathematical architecture: projective state space, symplectic structure, invariant measure, and Hamiltonian flow. CSD preserves that operational geometry while adding a realist interpretation beneath it. In the present framework, the ontic manifold Σ is taken to be physically real, the microstate ω is taken to be definite, and measurement is interpreted as the deterministic revelation of an outcome region rather than as probabilistic collapse.

On that basis, the paper has proposed a unified geometric account of several issues that standard formulations leave open. It has treated preparation as restriction to an ontic region, measurement as context-defined partitioning, decoherence as loss of isolation, and Born-form probabilities as the operational image of invariant geometric weight structure. The aim has not been to replace the successful operational content of quantum mechanics, but to provide a more explicit account of what that content is about.

In this sense, the present work should be understood as an extension rather than a replacement of GQM. The mathematical structure is inherited from the geometric formulation. What changes is the physical interpretation assigned to that structure.

11.2 Key Advantages of CSD

Relative to standard geometric quantum mechanics, CSD aims to provide the following conceptual gains:

- it introduces an explicit realist ontology in which the ontic microstate ω is definite and physically meaningful;
- it gives a deterministic account of measurement in terms of outcome regions on Σ;
- it clarifies the epistemic role of ψ and the geometric basis of outcome weights;
- it turns a powerful geometric reformulation into a candidate physical interpretation.

Relative to other major interpretations, CSD is intended to occupy a distinctive position:

- it retains the geometric mathematics of the standard quantum formalism while avoiding collapse postulates and branching ontologies;
- it is ontologically simpler than pilot-wave dual-state models because it does not posit a separate guiding wave as an additional physical entity;
- it is more definite than Everettian branching accounts because it maintains one continuous ontic manifold rather than many simultaneously realised worlds;
- it is more explicit than Copenhagen-style views because it attempts to explain definite outcomes rather than treating them as primitive.

These are intended as conceptual advantages, not as claims of immediate empirical superiority within the finite-dimensional regime studied here.

11.3 Looking Forward

The most productive future for this programme is likely to be collaborative rather than oppositional. Geometric quantum mechanics remains the essential mathematical foundation for the work, and its further development in differential geometry, symplectic structure, and quantum information will continue to sharpen the formal basis on which CSD depends. The contribution of CSD is to ask what follows if that same structure is taken ontologically rather than merely operationally.

Several next steps follow naturally from the present paper:

- further clarification and publication of the supporting technical notes and companion papers;

- formalisation of the core probability and measurement results;
- deeper analysis of the relation between the ontic manifold Σ and the projective operational description;
- extension toward larger systems, effective descriptions, and eventually field-theoretic or spacetime-level structure.

The long-term ambition is a synthesis in which the mathematical elegance of GQM and the ontological clarity sought by CSD reinforce one another.

11.4 Final Thought

Geometric quantum mechanics showed that quantum theory *can* be formulated geometrically. Constraint-Surface Dynamics advances the stronger claim that, within the present finite-dimensional scope, quantum theory may be most intelligible when understood through that geometry as physically meaningful rather than merely formal.

Whether this geometric ontology is ultimately only a useful interpretive reorganisation or part of the deeper structure of reality remains an open question. The present paper does not settle that question. It argues that the question is scientifically legitimate, mathematically well posed, and worth pursuing.

Scope note: The framework developed here is a deterministic geometric account of finite-dimensional quantum theory in the quantum-effective sector. Detailed derivations of broader Hamiltonian, relativistic, or field-theoretic structure are deferred to later work.

A Mathematical Notation

The mathematical core of CSD rests on the identification of a finite-dimensional ontic manifold Σ equipped with symplectic structure and, in the quantum-effective sector, a compatible projective geometric description. The purpose of this appendix is to fix notation and to collect the main geometric objects used throughout the paper.

A.1 The Constraint Surface Σ

Let Σ denote the ontic configuration manifold of the physical system. Each point

$$\omega \in \Sigma$$

represents a complete ontic microstate of the system, specifying all physical degrees of freedom accessible within the scope of the model.

Mathematically, Σ is taken to be a finite-dimensional smooth symplectic manifold

$$(\Sigma, \omega_\Sigma),$$

where ω_Σ is a closed, non-degenerate two-form. This symplectic structure determines Hamiltonian evolution.

Given a Hamiltonian function

$$H : \Sigma \to \mathbb{R},$$

the associated Hamiltonian vector field X_H is defined by

$$\iota_{X_H} \omega_\Sigma = dH.$$

The resulting flow

$$\phi_t : \Sigma \to \Sigma$$

gives the deterministic evolution of the ontic microstate:

$$\frac{d\omega}{dt} = X_H(\omega), \qquad \omega(t) = \phi_t(\omega_0).$$

In local Darboux coordinates (q^i, p_i), the symplectic form takes the canonical form

$$\omega_\Sigma = \sum_i dq^i \wedge dp_i.$$

The corresponding Liouville measure μ_Σ is preserved under Hamiltonian evolution. Equivalently, for every measurable region $\Omega \subset \Sigma$,

$$\mu_\Sigma(\phi_t(\Omega)) = \mu_\Sigma(\Omega).$$

This is the measure-preserving content of Liouville's theorem and underlies the later construction of geometric outcome weights.

A.2 Complex and Metric Structure

At the operational level, the finite-dimensional quantum-effective sector is represented on the complex projective manifold

$$\mathbb{CP}^{n-1},$$

equipped with its standard Kähler structure. This includes:

- the Fubini–Study symplectic form ω_{FS},
- the compatible complex structure J,
- and the Fubini–Study metric g_{FS}.

These structures satisfy the standard Kähler compatibility relation

$$g_{\mathrm{FS}}(X,Y)=\omega_{\mathrm{FS}}(X,JY), \qquad J^2=-\mathbb{I}.$$

Locally, one may introduce complex coordinates by combining canonical variables into expressions of the form

$$\psi^i=\frac{1}{\sqrt{2}}\left(q^i+ip_i\right),$$

so that the local geometry admits a complex representation. At the projective level, physical pure states correspond to rays, and the operational state space is therefore the projective quotient rather than the full complex coordinate space.

The Fubini–Study line element is induced from the Hilbert-space inner product and may be written schematically as

$$ds^2=4\Big(\langle d\psi|d\psi\rangle-|\langle\psi|d\psi\rangle|^2\Big),$$

which is invariant under the natural unitary action on projective state space.

In CSD, this projective geometry is not itself fundamental. Rather, it is the epistemic image of the deeper ontic dynamics on Σ, linked by the projection map

$$\pi:\Sigma\to\mathbb{CP}^{n-1}.$$

The manifold Σ is ontic, while $\mathbb{CP}^{n-1}$ carries the induced operational geometry relevant for the standard finite-dimensional quantum description.

A.3 The Invariant Measure

The operationally relevant measure in the finite-dimensional quantum-effective sector must be invariant under the symmetries of the projective state space. For systems with internal $\mathrm{SU}(n)$ symmetry, the unique measure satisfying

1. projective invariance, so that physical states are identified up to overall complex phase, and
2. covariance under the natural $\mathrm{SU}(n)$ action on projective state space,

is the Fubini–Study measure

$$\mu_{\mathrm{FS}}$$

on the projective manifold

$$\mathbb{CP}^{n-1}.$$

This measure provides the natural operational volume form used to define outcome weights in the projective description. If a measurement context induces outcome regions Ω_i relative to a preparation domain Ω_0, then the associated operational weight is

$$w_i=\frac{\mu_{\mathrm{FS}}(\Omega_i)}{\mu_{\mathrm{FS}}(\Omega_0)}.$$

In the standard projective representation, and after imposition of the admissible probability functional in the quantum-effective sector as developed in the companion measure-analysis, these weights take the familiar Born form

$$w_i=|\langle\phi_i\mid\Psi\rangle|^2.$$

Within CSD, this projective measure is not treated as an arbitrary probabilistic postulate. Rather, it is the unique invariant operational measure associated with the geometry of projective state space, and it is linked to the deeper ontic measure on Σ through the projection map

$$\pi:\Sigma\to\mathbb{CP}^{n-1}.$$

B Worked Example: The Qubit

The qubit provides the simplest nontrivial system on which the geometric construction of CSD can be displayed explicitly. A full derivation is given elsewhere. The present appendix records only the core geometric facts needed for the main text.

B.1 Geometric Setup

A two-level system has local coordinate description in $\mathbb{C}^2$, and its operational projective state space is the complex projective line

$$\mathbb{CP}^1 \cong S^2,$$

that is, the Bloch sphere. Each point on this sphere represents a distinct ray, while antipodal points represent orthogonal pure states.

The unique $\mathrm{SU}(2)$-invariant geometry on $\mathbb{CP}^1$ is the Fubini–Study geometry. In spherical coordinates (α, ϕ) on the Bloch sphere, the corresponding normalised invariant measure is

$$d\mu_{\mathrm{FS}} = \frac{\sin\alpha}{4\pi}\, d\alpha\, d\phi.$$

This measure provides the natural operational volume form used to compute geometric outcome weights as relative surface areas.

B.2 State and Measurement Basis

A general qubit preparation relative to a measurement basis $\{|\phi_0\rangle, |\phi_1\rangle\}$ may be written as

$$|\Psi\rangle = \cos\left(\frac{\theta}{2}\right)|\phi_0\rangle + e^{i\varphi}\sin\left(\frac{\theta}{2}\right)|\phi_1\rangle,$$

where θ is the Bloch-sphere angle between the preparation ray $[\Psi]$ and the eigenstate ray $[\phi_0]$.

The Fubini–Study distance between these rays is

$$d_{\mathrm{FS}}([\Psi], [\phi_0]) = \frac{\theta}{2},$$

so the geometric separation on projective state space directly encodes the amplitude overlap structure of the qubit.

B.3 Outcome Weights from Invariant Volumes

Because the $\mathrm{SU}(2)$ action on $\mathbb{CP}^1$ is transitive and the invariant measure is unique, the operational outcome regions for a qubit are determined purely by the angular geometry of the Bloch sphere. For the measurement basis $\{|\phi_0\rangle, |\phi_1\rangle\}$ and preparation state $|\Psi\rangle$ separated by Bloch angle θ, the corresponding invariant volume fractions yield the outcome weights

$$w_0 = \cos^2\left(\frac{\theta}{2}\right), \qquad w_1 = \sin^2\left(\frac{\theta}{2}\right).$$

These coincide exactly with the standard Born-rule probabilities for a qubit. In the geometric picture, they arise from invariant projective volume alone, without the addition of any stochastic collapse postulate.

B.4 Physical Meaning

The qubit case illustrates the central ideas of the framework in their simplest form:

- **Volume typicality:** Long-run outcome frequencies are associated with invariant surface-area ratios on $\mathbb{CP}^1$.
- **Symmetry constraint:** $\mathrm{SU}(2)$ invariance fixes the Fubini–Study measure uniquely, so no adjustable measure choice remains.
- **Calibration role:** The qubit is the minimal nontrivial consistency test of the geometric construction. Any deterministic volume-based framework intended to reproduce Born-form probabilities must recover these weights in the two-level case.

The result confirms that deterministic, measure-preserving geometric structure reproduces the empirical $|\psi|^2$ law for two-level systems at the operational level.

B.5 Extension to Higher Dimensions

The same basic reasoning extends beyond the qubit. On

$$\mathbb{CP}^{n-1},$$

the combined requirements of projective invariance and $\mathrm{SU}(n)$ covariance uniquely select the Fubini–Study measure as the canonical operational measure. In higher dimensions, however, the recovery of Born-form probabilities requires not only the invariant measure but also the admissible operational probability functional. The resulting Born weights therefore arise through the same modular chain used throughout this paper: invariant geometric measure, operational admissibility, and the typicality-to-frequency link.

The qubit case is therefore both an explicit worked demonstration and a benchmark example. It shows concretely how the volume-based framework operates in the simplest nontrivial setting, while the higher-dimensional generalisation supplies the broader operational scope relevant to finite-dimensional quantum theory.

The explicit construction of measurement partitions in higher-dimensional projective spaces is treated elsewhere in the programme.

References

Fabio Anza and James P. Crutchfield. Geometric quantum information measures. *PRX Quantum*, 3(2):020310, May 2022. ISSN 2691-3399. doi:10.1103/PRXQuantum.3.020310. URL `http://dx.doi.org/10.1103/PRXQuantum.3.020310`.

Abhay Ashtekar and Troy A. Schilling. Geometrical formulation of quantum mechanics. In A. Harvey, editor, *On Einstein's Path*, pages 23–65. Springer New York, 1999. ISBN 9781461214229. doi:10.1007/978-1-4612-1422-9_3. URL `http://dx.doi.org/10.1007/978-1-4612-1422-9_3`.

Guido Bacciagaluppi. *The Role of Decoherence in Quantum Mechanics*. Elements in Physics. Cambridge University Press, December 2020. ISBN 9781108774208. doi:10.1017/9781108774208. URL `http://dx.doi.org/10.1017/9781108774208`.

Ingemar Bengtsson and Karol Życzkowski. *Geometry of Quantum States: An Introduction to Quantum Entanglement*. Cambridge University Press, May 2006. ISBN 9780511535048. doi:10.1017/CBO9780511535048. URL `http://dx.doi.org/10.1017/CBO9780511535048`.

Zayn Blore. Volume-based probability: Outcome frequencies from deterministic geometry. 2025a. doi:10.5281/zenodo.15564079. URL `https://doi.org/10.5281/zenodo.15564079`.

Zayn Blore. Fixing the measure: Symmetry and the geometric origin of the born rule. 2025b. doi:10.5281/zenodo.16746829. URL `https://doi.org/10.5281/zenodo.16746829`.

Zayn Blore. A deterministic reconstruction of finite-dimensional quantum mechanics. 2025c. doi:10.5281/zenodo.18098064. URL `https://doi.org/10.5281/zenodo.18098064`.

David Bohm. A suggested interpretation of the quantum theory in terms of "hidden" variables. I. *Physical Review*, 85(2):166–179, January 1952a. ISSN 1536-6065. doi:10.1103/PhysRev.85.166. URL `http://dx.doi.org/10.1103/PhysRev.85.166`.

David Bohm. A suggested interpretation of the quantum theory in terms of "hidden" variables. II. *Physical Review*, 85(2):180–193, January 1952b. ISSN 1536-6065. doi:10.1103/PhysRev.85.180. URL `http://dx.doi.org/10.1103/PhysRev.85.180`.

Heinz-Peter Breuer and Francesco Petruccione. *The Theory of Open Quantum Systems*. Oxford University Press, January 2002. ISBN 9780198520634.

D. C. Brody and L. P. Hughston. Geometric quantum mechanics. *Journal of Geometry and Physics*, 38(1):19–53, April 2001. ISSN 0393-0440. doi:10.1016/S0393-0440(00)00052-8. URL `http://dx.doi.org/10.1016/S0393-0440(00)00052-8`.

R. Cirelli, A. Mania, and L. Pizzocchero. Quantum mechanics as an infinite-dimensional hamiltonian system with uncertainty structure: Part i. *Journal of Mathematical Physics*, 31(10):2891–2897, October 1990. ISSN 1089-7658. doi:10.1063/1.528801. URL `http://dx.doi.org/10.1063/1.528801`.

Giacomo Mauro D'Ariano, Giulio Chiribella, and Paolo Perinotti. *Quantum Theory from First Principles*. Cambridge University Press, April 2017. ISBN 9781107338340. doi:10.1017/9781107338340. URL `http://dx.doi.org/10.1017/9781107338340`.

Hugh Everett. "relative state" formulation of quantum mechanics. *Reviews of Modern Physics*, 29(3):454–462, July 1957. ISSN 1539-0756. doi:10.1103/RevModPhys.29.454. URL http://dx.doi.org/10.1103/RevModPhys.29.454.

Paolo Facchi, Ugo Marzolino, Giorgio Parisi, Saverio Pascazio, and Antonello Scardicchio. Phase transitions of bipartite entanglement. *Journal of Physics A: Mathematical and Theoretical*, 43(22):225303, May 2010. ISSN 1751-8121. doi:10.1088/1751-8113/43/22/225303. URL http://dx.doi.org/10.1088/1751-8113/43/22/225303.

Arthur Fine. Hidden variables, joint probability, and the bell inequalities. *Physical Review Letters*, 48(5):291–295, February 1982. ISSN 1079-7114. doi:10.1103/PhysRevLett.48.291. URL http://dx.doi.org/10.1103/PhysRevLett.48.291.

Christopher A. Fuchs and Rüdiger Schack. Quantum-bayesian coherence. *Reviews of Modern Physics*, 85(4):1693–1715, December 2013. ISSN 1539-0756. doi:10.1103/RevModPhys.85.1693. URL http://dx.doi.org/10.1103/RevModPhys.85.1693.

G. C. Ghirardi, A. Rimini, and T. Weber. Unified dynamics for microscopic and macroscopic systems. *Physical Review D*, 34(2):470–491, July 1986. ISSN 1550-7998. doi:10.1103/PhysRevD.34.470. URL http://dx.doi.org/10.1103/PhysRevD.34.470.

V. Gorini, A. Kossakowski, and E. C. G. Sudarshan. Completely positive dynamical semigroups of N-level systems. *Journal of Mathematical Physics*, 17(5):821–825, May 1976. ISSN 1089-7658. doi:10.1063/1.522979. URL http://dx.doi.org/10.1063/1.522979.

Lucien Hardy. Quantum theory from five reasonable axioms, January 2001. URL https://arxiv.org/abs/quant-ph/0101012.

Lucien Hardy. Reformulating quantum theory in terms of operational postulates, April 2011. URL https://arxiv.org/abs/1104.2066.

Nicholas Harrigan and Robert W. Spekkens. Einstein, incompleteness, and the epistemic view of quantum states. *Foundations of Physics*, 40(2):125–157, February 2010. ISSN 1572-9516. doi:10.1007/s10701-009-9347-0. URL http://dx.doi.org/10.1007/s10701-009-9347-0.

E. Joos, H. D. Zeh, C. Kiefer, D. J. W. Giulini, J. Kupsch, and I.-O. Stamatescu. *Decoherence and the Appearance of a Classical World in Quantum Theory*. Springer Berlin Heidelberg, 2003. ISBN 9783540003908. doi:10.1007/978-3-540-00390-8. URL http://dx.doi.org/10.1007/978-3-540-00390-8.

T. W. B. Kibble. Geometrization of quantum mechanics. *Communications in Mathematical Physics*, 65(2):189–201, June 1979. ISSN 1432-0916. doi:10.1007/BF01940902. URL http://dx.doi.org/10.1007/BF01940902.

Simon Kochen and Ernst P. Specker. The problem of hidden variables in quantum mechanics. *Journal of Mathematics and Mechanics*, 17(1):59–87, 1967.

Matthew S. Leifer. Is the quantum state real? an extended review of ψ-ontology theorems. *Quanta*, 3(1):67–155, 2014. ISSN 2161-3342. doi:10.12743/quanta.v3i1.22. URL http://dx.doi.org/10.12743/quanta.v3i1.22.

G. Lindblad. On the generators of quantum dynamical semigroups. *Communications in Mathematical Physics*, 48(2):119–130, June 1976. ISSN 1432-0916. doi:10.1007/BF01608499. URL http://dx.doi.org/10.1007/BF01608499.

J. P. Provost and G. Vallée. Riemannian structure on manifolds of quantum states. *Communications in Mathematical Physics*, 76(3):289–301, September 1980. ISSN 1432-0916. doi:10.1007/BF02193559. URL http://dx.doi.org/10.1007/BF02193559.

Matthew F. Pusey, Jonathan Barrett, and Terry Rudolph. On the reality of the quantum state. *Nature Physics*, 8(6):475–478, May 2012. ISSN 1745-2481. doi:10.1038/nphys2309. URL http://dx.doi.org/10.1038/nphys2309.

Carlo Rovelli. Relational quantum mechanics. *International Journal of Theoretical Physics*, 35(8):1637–1678, August 1996. ISSN 1572-9575. doi:10.1007/BF02302261. URL http://dx.doi.org/10.1007/BF02302261.

Maximilian Schlosshauer. Decoherence, the measurement problem, and interpretations of quantum mechanics. *Reviews of Modern Physics*, 76(4):1267–1305, January 2005. ISSN 1539-0756. doi:10.1103/RevModPhys.76.1267. URL http://dx.doi.org/10.1103/RevModPhys.76.1267.

William K. Wootters. Statistical distance and hilbert space. *Physical Review D*, 23(2):357–362, January 1981. ISSN 1550-7998. doi:10.1103/PhysRevD.23.357. URL http://dx.doi.org/10.1103/PhysRevD.23.357.

Wojciech H. Zurek. Decoherence and the transition from quantum to classical. *Physics Today*, 44(10):36–44, October 1991. ISSN 1945-0699. doi:10.1063/1.881293. URL http://dx.doi.org/10.1063/1.881293.

Wojciech H. Zurek. Decoherence, einselection, and the quantum origins of the classical. *Reviews of Modern Physics*, 75(3):715–775, May 2003. ISSN 1539-0756. doi:10.1103/RevModPhys.75.715. URL http://dx.doi.org/10.1103/RevModPhys.75.715.